MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

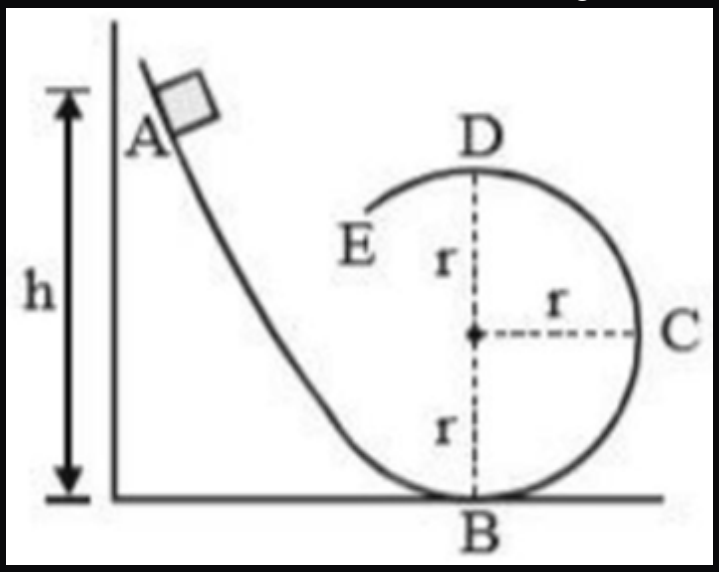
## GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
**Section 1:** Single Correct Type Questions (4 options, only one correct)
**Section 2:** Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
**Single Correct:** +4 for correct, -1 for wrong, 0 if unattempted.
**Numerical Value:** +4 for correct, -1 for wrong.

By: C I P H Ξ R

**Q.1**

ABCDE is a channel in the vertical plane, part $BCDE$ being circular with radius $r$. A block is released from $A$ and slides without friction and without rolling. The block will complete the loop if $h$ is :-




**(A)** $h \leq \frac{3}{2}r$

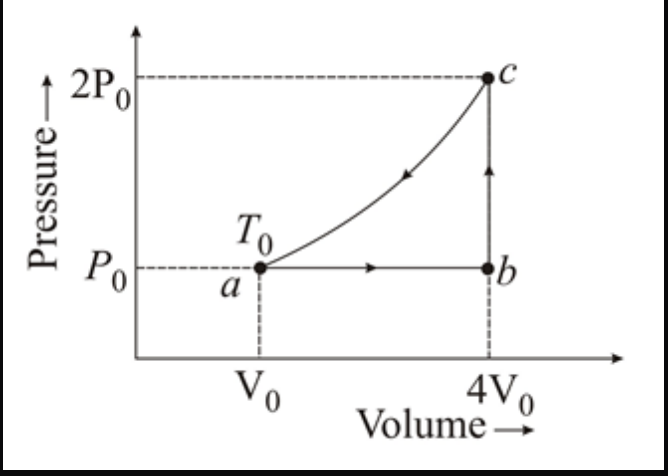
**(B)** $h \geq \frac{5}{2}r$

**(C)** $h \geq \frac{3}{2}r$

**(D)** $h \leq \frac{5}{2}r$

**Q.2**

One mole of an ideal monoatomic gas is made to go through the cycle as shown in figure. Then the change in the internal energy in expanding the gas from *a* to *c* along the path *abc* is




**(A)** $3\mathrm{P}_0\mathrm{V}_0$

**(B)** $6\,\mathrm{RT}_0$

**(C)** $4.5RT_0$

**(D)** $10.5\,\mathrm{RT}_0$

**Q.3**

Three closed vessels $A$, $B$ and $C$ are at the same temperature $\mathrm{T}$ and contain gases which obey the Maxwellian distribution of velocities. Vessel A contains only $\mathrm{O}_2$, $B$ only $\mathrm{N}_2$ and $\mathrm{C}$ a mixture of equal quantities of $\mathrm{O}_2$ and $N_2$. If the average speed of the $O_2$ molecules in vessel $A$ is $V_1$, that of the $N_2$ molecules in vessel $B$ is $V_2$, the average speed of the $O_2$ molecules in vessel $C$ is

**(A)** $(V_1 + V_2)/2$

**(B)** $V_1$

**(C)** $(V_1V_2)^{1/2}$

**(D)** $\sqrt{3kT/M}$

**Q.4**

Electrons in a certain energy level $n = n_1$, can emit $3$ spectral lines. When they are in another energy level, $\mathrm{n} = \mathrm{n}_2$ . They can emit $6$ spectral lines. The orbital speed of the electrons in the two orbits are in the ratio of

**(A)** $4:3$

**(B)** $3:4$

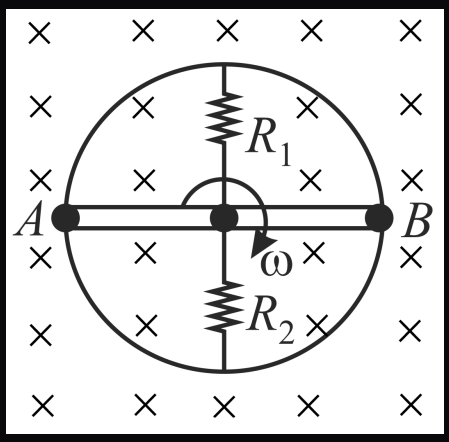
**(C)** $2:1$

**(D)** $1:2$

**Q.5**

AB is a resistance less conducting rod which forms a diameter of a conducting ring of radius r rotating in a uniform magnetic field B as shown in the diagram. The resistors $\mathrm{R}_1$ and $\mathrm{R}_2$ do not rotate. Then the current through the resistor

$R_1$ is




**(A)** $\frac{B\omega r^2}{2R_1}$

**(B)** $\frac{B\omega r^2}{2R_2}$

**(C)** $\frac{B\omega r^2}{2R_1R_2}(R_1+R_2)$

**(D)** $\frac{B\omega r^2}{2(R_1+R_2)}$

**Q.6**

An open carriage in a goods train is moving with a uniform velocity of 10 $ms^{-1}$. If the rain adds water with zero velocity at the rate of 5 $kgs^{-1}$, then the additional force applied by the engine to maintain the same velocity of the train is

**(A)** 0.5 N

**(B)** 2.0 N

**(C)** 50 N

**(D)** 25 N

**Q.7**

When an object is shot from the bottom of a long smooth inclined plane kept at an angle $60^\circ$ with horizontal, it can travel a distance $x_1$ along the plane. But, when the inclination is decreased to $30^\circ$ and the same object is shot with the same velocity, it can travel $x_2$ distance. Then $x_1 : x_2$ will be

**(A)** $1:\sqrt{2}$

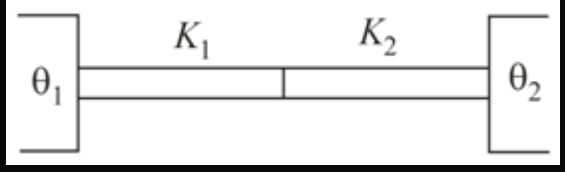
**(B)** $\sqrt{2}:1$

**(C)** $1:\sqrt{3}$

**(D)** $1:2\sqrt{3}$

**Q.8**

Two identical rods are made of different materials whose thermal conductivities are $K_1$ and $K_2$. They are placed end to end between two heat reservoirs at temperatures $\theta_1$ and $\theta_2$. The temperature of the junction of the rod is




**(A)** $\frac{\theta_1+\theta_2}{2}$

**(B)** $\frac{K_1\theta_1+K_2\theta_2}{K_1+K_2}$

**(C)** $\frac{K_1\theta_2+K_2\theta_1}{K_1+K_2}$

**(D)** $\frac{K_1\theta_1+K_2\theta_2}{|K_1-K_2|}$

**Q.9**

Stationary waves of frequency $200\ \text{Hz}$ are formed in air. If the velocity of the wave is $360\ \text{m/s}$, the shortest distance between two anti-nodes is

**(A)** $1.8\ \text{m}$

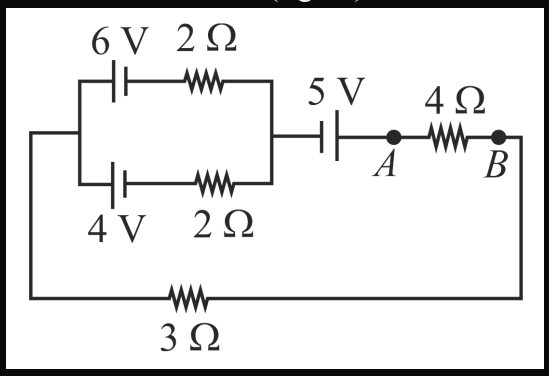
**(B)** $3.6\ \text{m}$

**(C)** $0.9\ \text{m}$

**(D)** $0.45\ \text{m}$

**Q.10**

In the circuit shown (figure), calculate the current through 6 V battery.




**(A)** $(1/4)\ \text{A}$

**(B)** $(1/8)\ \text{A}$

**(C)** $(1/2)\ \text{A}$

**(D)** None of these

**Q.11**

Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole $(\text{d} << \text{R})$. The value of d is

**(A)** $\frac{\omega^2\text{R}^2}{\text{g}}$

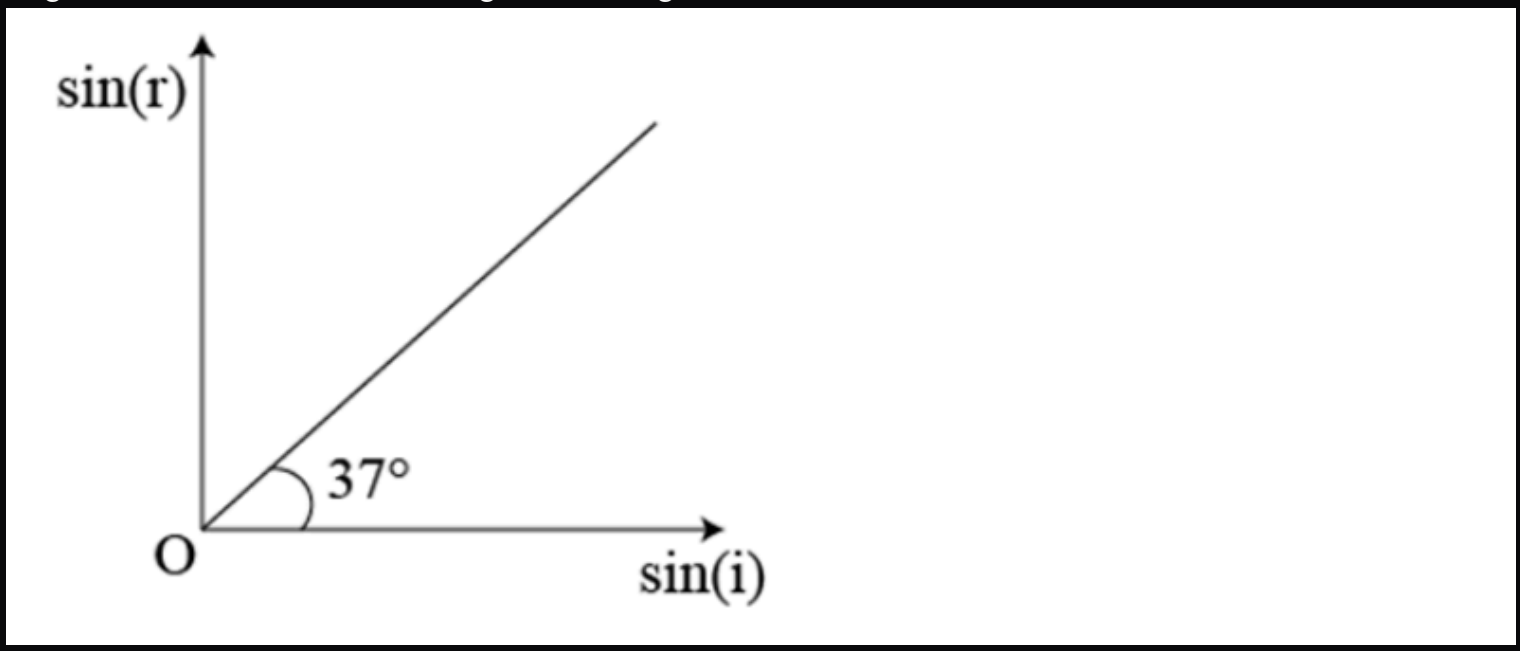
**(B)** $\frac{\omega^2 R^2}{2g}$

**(C)** $\frac{2\omega^2 R^2}{g}$

**(D)** $\frac{\omega^2 R^2}{4g}$

**Q.12**

A ray of light, travelling in medium $A$, is incident on plane interface of two media $A$ and $B$ and gets refracted into medium $B$. The angle of incidence is $i$ and that of refraction is $r$. Graph between $\sin(i)$ and $\sin(r)$ is as shown in diagram. The correct statement among the following is




**(A)** Speed of light in medium $B$ is three fourth of that in medium $A$

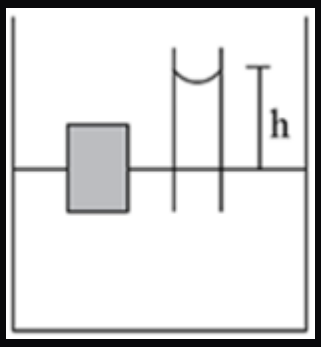
**(B)** Total internal reflection can take place

**(C)** Refraction index of medium $A$ is greater than that of medium $B$

**(D)** None of these

**Q.13**

In a capillary tube placed inside the liquid of density ' $\delta'$ in a container, the rise of liquid is $h$. When block of density ' $\sigma'$ is placed on the liquid as shown in the figure, the liquid rise in the tube is $h'$, if $\sigma < \delta$ then :




**(A)** $h' = h$

**(B)** $h' < h$

**(C)** $h' > h$

**(D)** insufficient data

**Q.14**

If the energy, $E = G^p h^q c^r$, where $G$ is the universal gravitational constant, $h$ is the Planck's constant and $c$ is the velocity of light, then the values of $p$, $q$ and $r$ are, respectively

**(A)** $-\frac{1}{2}, \frac{1}{2}$ and $\frac{5}{2}$

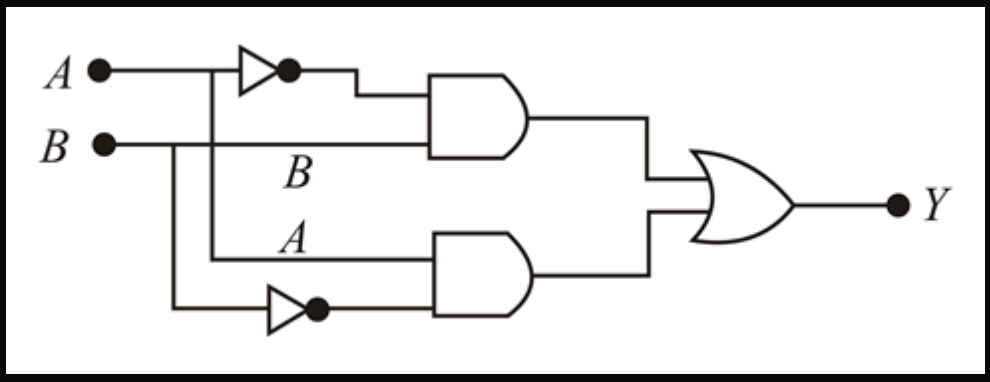
**(B)** $\frac{1}{2}, -\frac{1}{2}$ and $-\frac{5}{2}$

**(C)** $-\frac{1}{2}, \frac{1}{2}$ and $\frac{3}{2}$

**(D)** $\frac{1}{2}, -\frac{1}{2}$ and $-\frac{3}{2}$

**Q.15**

The truth table for the following logic circuit is




**(A)**

| $A$ | $B$ | $Y$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**(B)**

| $A$ | $B$ | $Y$ |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

**(C)**

| $A$ | $B$ | $Y$ |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**(D)**

| $A$ | $B$ | $Y$ |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

**Q.16**

A transformer with efficiency $80\%$ works at $4\ \mathrm{kW}$ and $100\ \mathrm{V}$. If the secondary voltage is $200\ \mathrm{V}$, then the primary and secondary currents are respectively

**(A)** $40\ \mathrm{A}$ and $16\ \mathrm{A}$

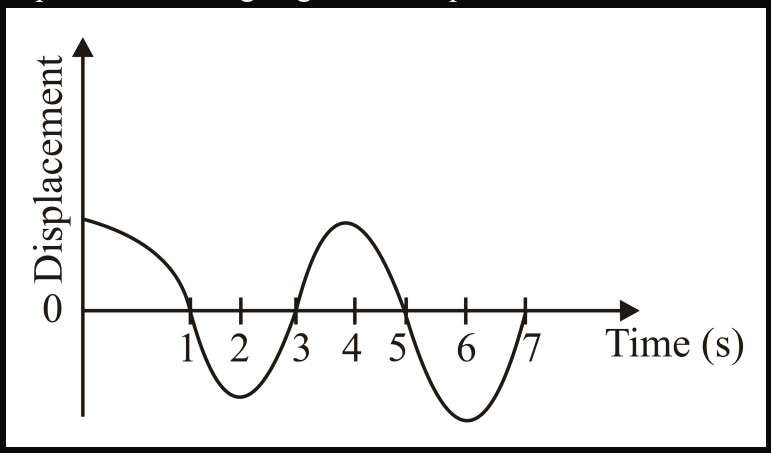
**(B)** $16\ \mathrm{A}$ and $40\ \mathrm{A}$

**(C)** $20\ \mathrm{A}$ and $90\ \mathrm{A}$

**(D)** $40\ \mathrm{A}$ and $20\ \mathrm{A}$

**Q.17**

A particle is undergoing SHM. Displacement vs time curve is as shown in the figure.




The correct statement is

**(A)** Phase of the oscillator is same at $\mathrm{t} = 0\ \mathrm{s}$ and $\mathrm{t} = 2\ \mathrm{s}$.

**(B)** Phase of the oscillator is same at $\mathrm{t} = 2\ \mathrm{s}$ and $\mathrm{t} = 5\ \mathrm{s}$.

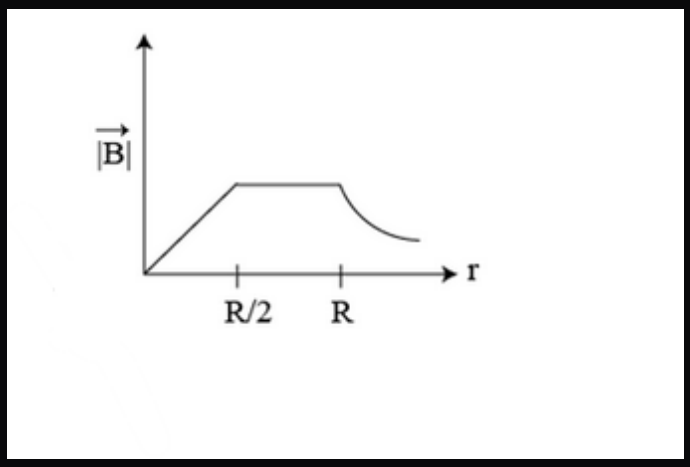
**(C)** Phase of the oscillator is same at $\mathrm{t} = 1\ \mathrm{s}$ and $\mathrm{t} = 7\ \mathrm{s}$.

**(D)** Phase of the oscillator is same at $\mathrm{t} = 1\ \mathrm{s}$ and $\mathrm{t} = 5\ \mathrm{s}$.

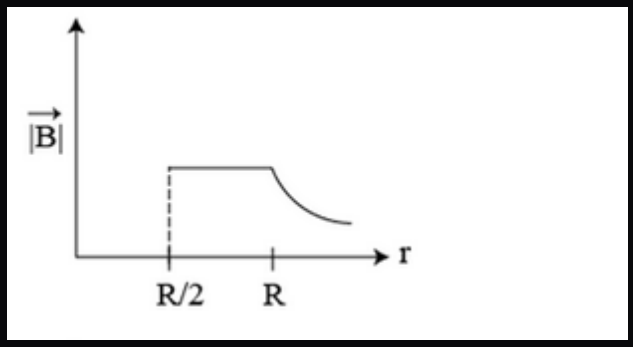
**Q.18**

An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius $R$ carries a uniform current density along its length. The magnitude of the magnetic field, $B$ as a function of the radial distance $r$ from the axis, is best represented by:

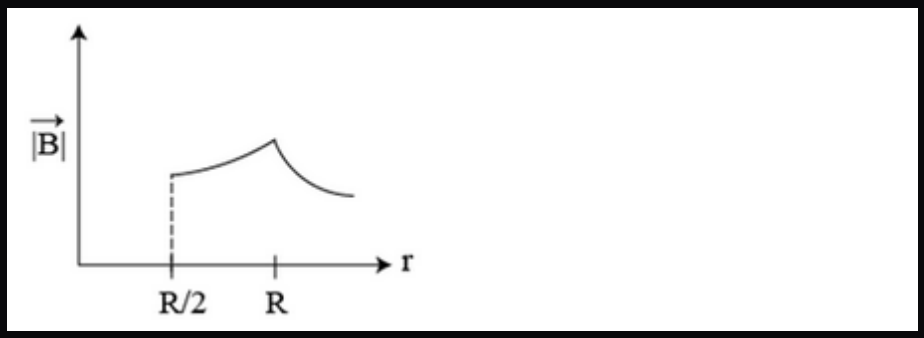
**(A)**



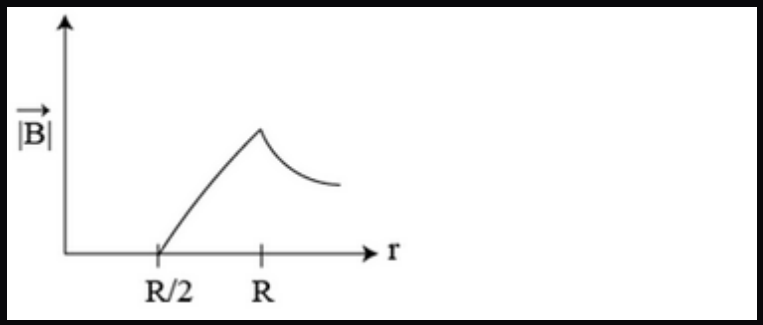

**(B)**




**(C)**



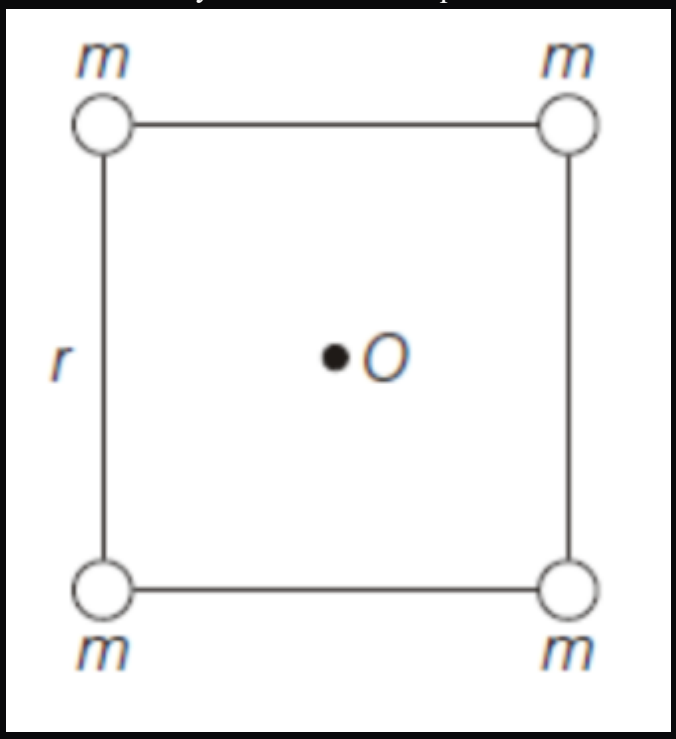

**(D)**




**Q.19**

There is a system of four mass particles placed (fixed) at the edge of a square of side $r$. The workdone to bring a mass

$m$ from infinity to the centre of square is




**(A)** $2\sqrt{2}Gm^2/r$

**(B)** $2\mathrm{Gm}^2/\mathrm{r}$

**(C)** $\frac{4\sqrt{2}Gm^2}{r}$

**(D)** $\frac{4Gm^2}{r}$

**Q.20**

The energy of a photon is equal to the kinetic energy of a proton. The energy of the photon is $E$. Let $\lambda_1$ be the de-Broglie wavelength of the proton and $\lambda_2$ be the wavelength of the photon. The ratio $\frac{\lambda_1}{\lambda_2}$ is proportional to

**(A)** $E^0$

**(B)** $E^{\frac{1}{2}}$

**(C)** $E^{-1}$

**(D)** $E^{-2}$

**Q.21**

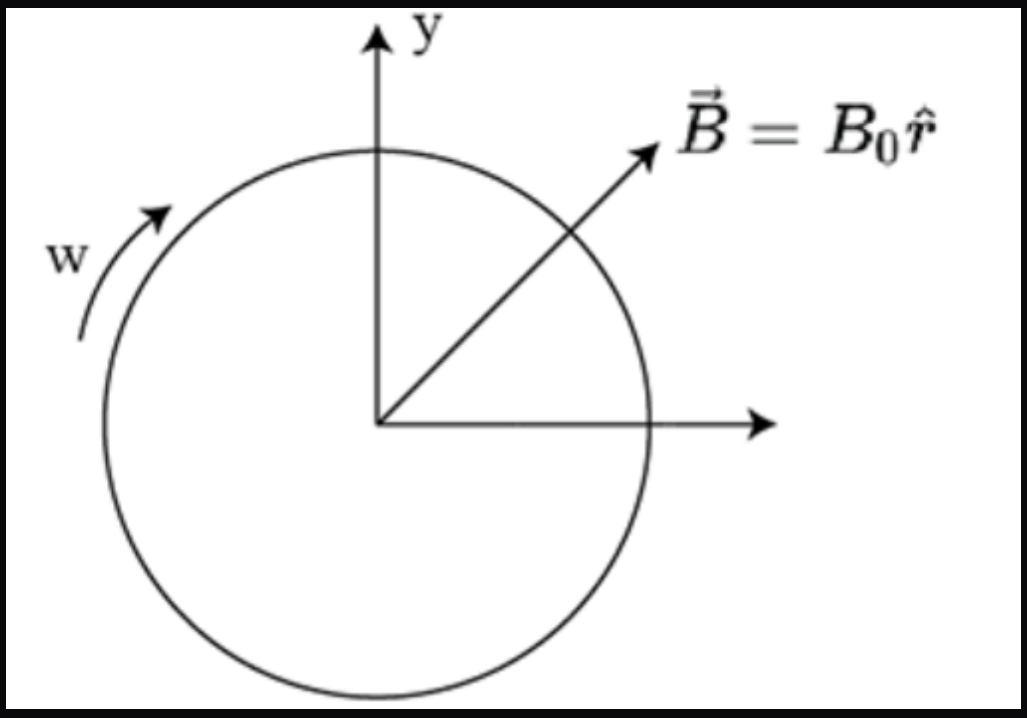
A hydrogen-like atom of atomic number $\mathrm{Z}$ is in an excited state of quantum number $2n$. It can emit a maximum energy photon of $204\mathrm{eV}$. If it makes a transition to the quantum state $n$, a photon of energy $40.8\mathrm{eV}$ is emitted. Calculate the atomic number $Z$. Ground state energy of hydrogen atom is $-13.6\mathrm{eV}$

**Q.22**

Two trains of equal lengths take 10 s and 15 s respectively to cross a telegraph post. If the length of each train be 120 m, in how much time (in s) will they cross each other travelling in opposite direction?

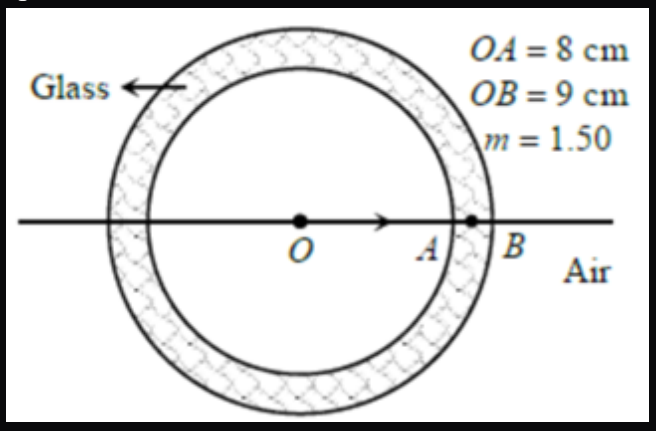
**Q.23**

A thin non conducting disc of mass $M = 2\ \text{kg}$, charge $Q = 2 \times 10^{-2}\text{C}$ and radius $R = \frac{1}{6}\ \text{m}$ is placed on a frictionless horizontal plane with its centre at the origin of the coordinate system. A nonuniform, radial magnetic field $\vec{B} = B_0\hat{r}$ exists in space, where $B_0 = 10\ \text{T}$ and $\hat{r}$ is a unit vector in the radially outward direction. The disc is set in motion with an angular velocity $\omega = x \times 10^2 \text{rads}^{-1}$, about an axis passing through its centre and perpendicular to its plane, as shown in the figure. At what value of $x$, the disc will lift off from the surface.




**Q.24**

A small filament is at the centre of a hollow glass sphere of inner and outer radii $8\ \text{cm}$ and $9\ \text{cm}$ respectively. The refractive index of glass is $1.50$. Calculate the position of the image of the filament when viewed from outside the sphere.

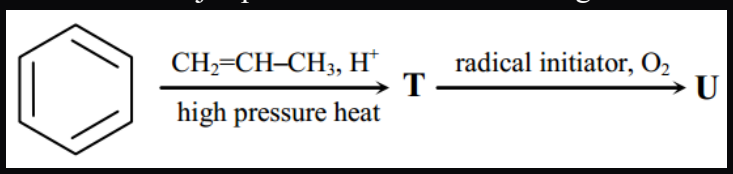



**Q.25**

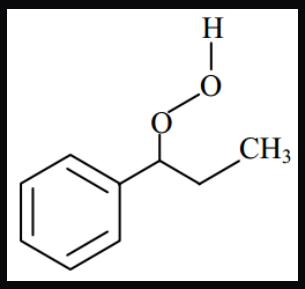
The mass and volume of a body are found to be $5.00\ \pm 0.05$ kg and $1.00\ \pm 0.05\ \text{m}^3$ respectively. Then the maximum possible percentage error in its density is

**Q.26**

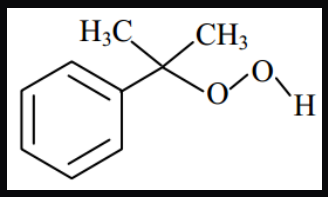
What is the major product 'U' in the following reactions?



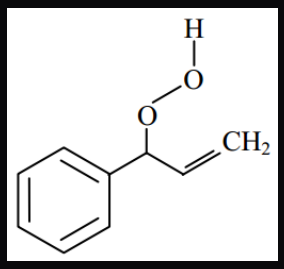
**(A)**



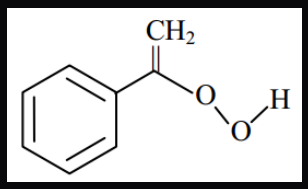

**(B)**




**(C)**




**(D)**




**Q.27**

A nucleoside on hydrolysis gives

**(A)** an aldopentose and a nitrogenous base

**(B)** an aldopentose and phosphoric acid

**(C)** an aldopentose, a nitrogenous base and phosphoric acid

**(D)** a nitrogenous base and phosphoric acid

**Q.28**

In a period, atom with smallest radius generally is:

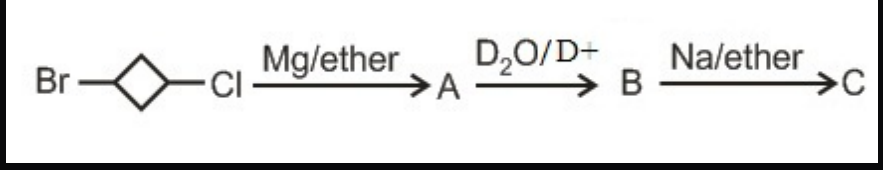
**(A)** Chalcogen

**(B)** Halogen

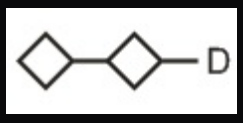
**(C)** Aerogen

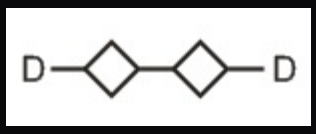
**(D)** Pnicogen

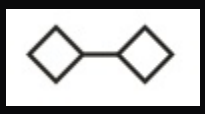
**Q.29**




C is

**(A)** 


**(B)** 


**(C)** 

**(D)** None of these

**Q.30**

The equilibrium constant for the given reaction is approximately $10^{-3}$

$$HPO_4^{2-}(aq) + HCO_3^-(aq) \rightleftharpoons H_2PO_4^-(aq) + CO_3^{2-}(aq)$$

Which is strongest conjugate base in the given reaction?

**(A)** $HPO_4^{2-}(aq)$

**(B)** $HCO_3^-(aq)$

**(C)** $H_2PO_4^-(aq)$

**(D)** $CO_3^{2-}(aq)$

**Q.31**

The compound which cannot be oxidised by $O_3$ gas is:

**(A)** $KI$

**(B)** $FeSO_4$

**(C)** $KMnO_4$

**(D)** $K_2MnO_4$

**Q.32**

Which of the following is a correct order of bond order ?

**(A)** $C_2 < NO < He_2^+ < O_2^-$

**(B)** $NO < C_2 < O_2^- < He_2^+$

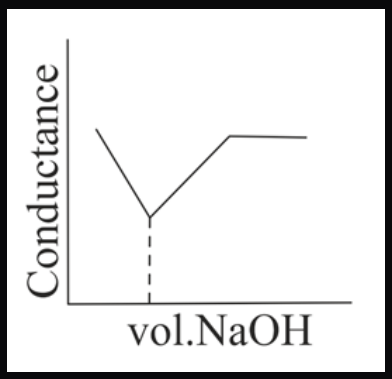
(C) $He_2^+ < O_2^- < NO < C_2$

(D) $He_2^+ < O_2^- < C_2 < NO$

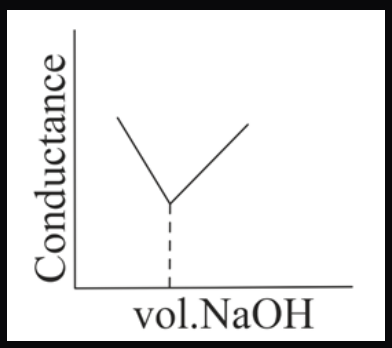
**Q.33**

Among the following plots, which plot will obtain for a conductometic titration of a equimolar mixture of HCl and HCN with $NaOH_{(aq)}$ solution?

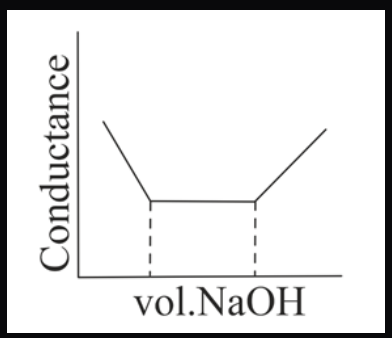
**(A)**



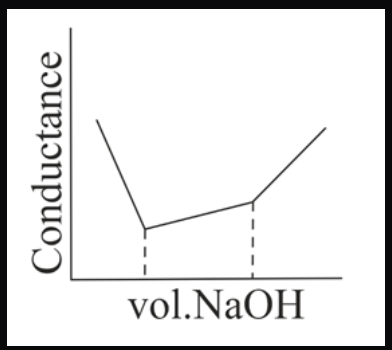

**(B)**




**(C)**




**(D)**




**Q.34**

The standard entropy of $X_2Y_2$ and $XY_3$ are 60, 40 and 50 $JK^{-1}$ $mol^{-1}$, respectively. For the reaction $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -30$ kJ, to be at equilibrium the temperature will be

(A) 1000 K

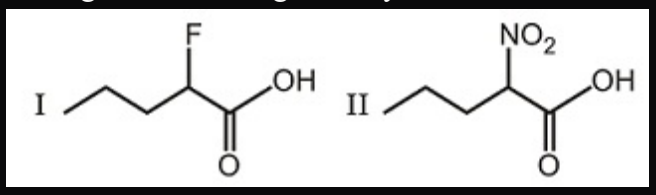
(B) 1250 K

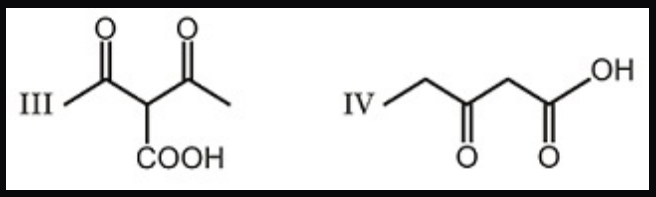
(C) 500 K

(D) 750 K

**Q.35**

Arrange the following carboxylic acids in increasing order of reactivity towards thermal decarboxylation reaction




(A) I < II < III < IV

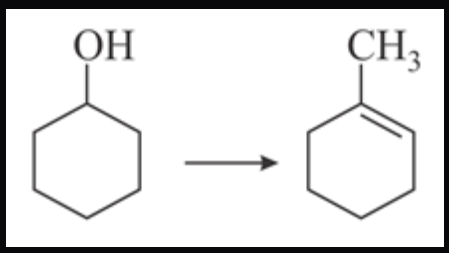
(B) I < III < IV < II

(C) I < IV < II < III

(D) IV < I < II < III

**Q.36**

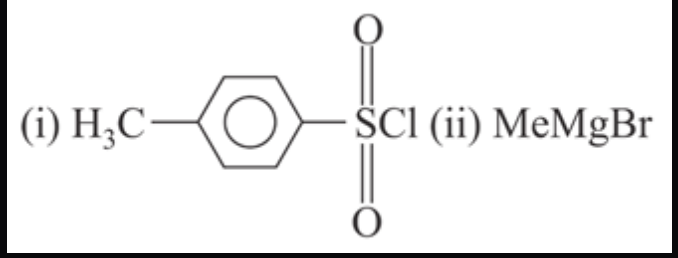
Which reaction condition would be best to perform the following transformation ?




(A)

(i) PCC, (ii) MeMgBr, (iii) $POCl_3$

(B)




(C)

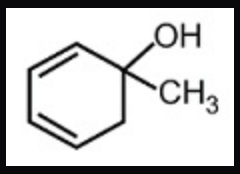
(i) HBr, (ii) Mg, (iii) MeI

(D)

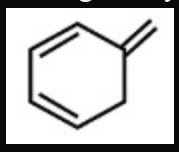
(i) NaOH, (ii) MeI

**Q.37**

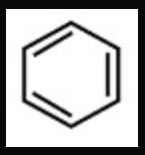
Statement I




undergo dehydration more easily to give

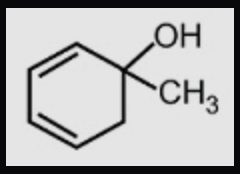


as a minor product and

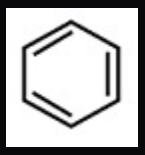


as a major product.

Statement II




is a non aromatic compound while



is an aromatic compound.

**(A)** Both statement I and statement II are true and the statement II is the correct explanation of the statement I

**(B)** Both statement I and statement II are true but the statement II is not the correct explanation of the statement I

**(C)** Statement I is true but statement II is false

**(D)** Both statement I and statement II are false

**Q.38**

The activation energy for a reaction at the temperature T K was found to be $2.303\,\mathrm{RT}$ J $\mathrm{mol}^{-1}$. The ratio of the rate constant to Arrhenius factor is

**(A)** $10^{-1}$

**(B)** $10^{-2}$

**(C)** $2 \times 10^{-3}$

**(D)** $2 \times 10^{-2}$

**Q.39**

If the vapour pressures of pure A and pure B at 298 K are 60 and 15 torr respectively, what would be the mole fraction of A in vapour phase (at this temperature) in a solution that contains 20% of A in (A + B) binary mixture in the liquid phase?

**(A)** 0.2

**(B)** 0.3

**(C)** 0.5

**(D)** 0.7

**Q.40**

Identify the product in the following reaction 3,4,5 - Tribromoaniline $\xrightarrow[(ii)H_3PO_2]{(i)\text{diazotization}}$ ?

**(A)** 3,4,5 - Tribromo nitroaniline

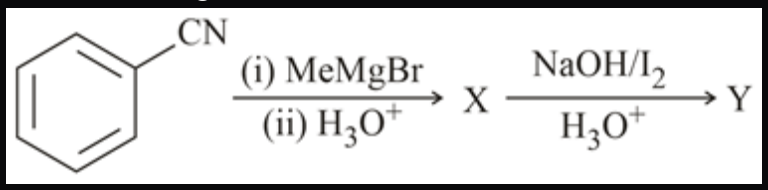
**(B)** 1,2,3 - Tribromobenzene

**(C)** 3,4,5 - Tribromophenol

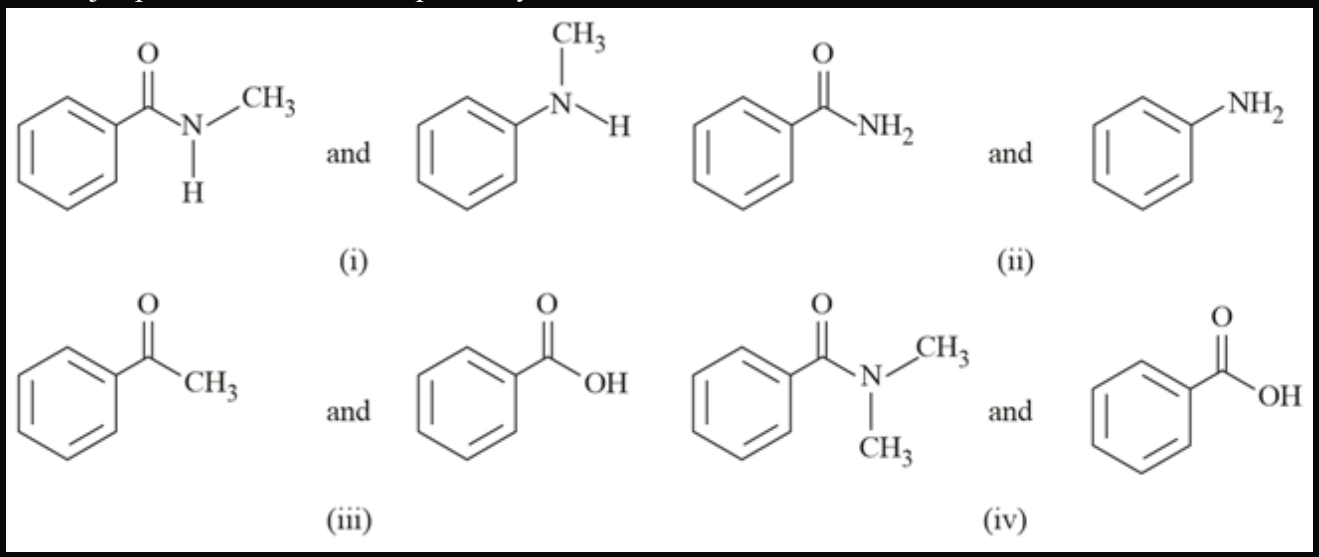
**(D)** 2,4,6 - Tribromobenzene

**Q.41**

In the following conversion,




the major products X and Y, respectively, are




**(A)** i

**(B)** ii

**(C)** iii

**(D)** iv

**Q.42**

Which among the given statements are true?

(i) $PH_5$ and $BiCl_5$ does not exist

(ii) $p\pi - d\pi$ bond is present in $SO_2$

(iii) $I_3^+$ has bent geometry according to VSEPR Theory

(iv) $SeF_4$ and $CH_4$ have same shape and geometry

**(A)** i, ii, iii

**(B)** i, iii

**(C)** i, iii, iv

**(D)** i, ii, iv

**Q.43**

Given the molecular formula of the hexacoordinated complexes (i) $CoCl_3 \cdot 6NH_3$, (ii) $CoCl_3.5NH_3$, (iii) $CoCl_3 \cdot 4NH_3$

If the number of co-ordinated $NH_3$ molecules in i, ii and iii respectively are $6, 5, 4$, the primary valencies in (i), (ii) and (iii) are :

**(A)** $6, 5, 4$

**(B)** $3, 2, 1$

**(C)** $0, 1, 2$

**(D)** $3, 3, 3$

**Q.44**

Which of the following reaction conditions is used for the conversion of ethanol to ethylene?

**(A)** conc. $H_2SO_4/70^\circ C$

**(B)** dil. $H_2SO_4/140^\circ C$

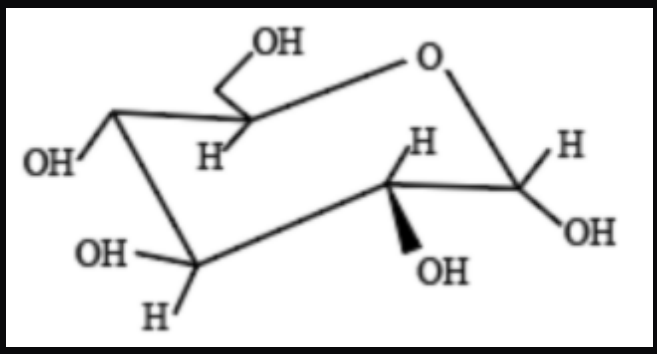
**(C)** dil. $H_2SO_4/100^\circ C$

**(D)** conc. $H_2SO_4/170^\circ C$

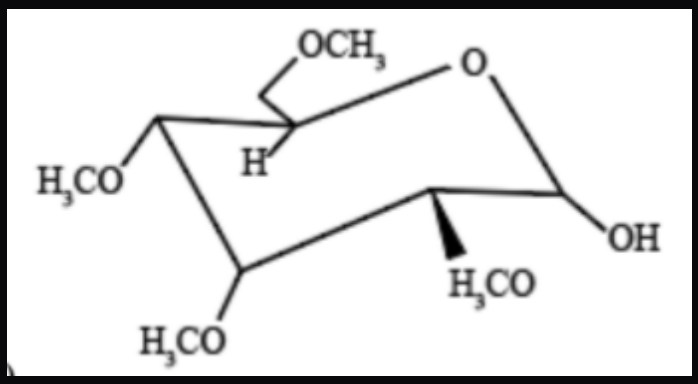
**Q.45**

Identify the non-reducing sugar.

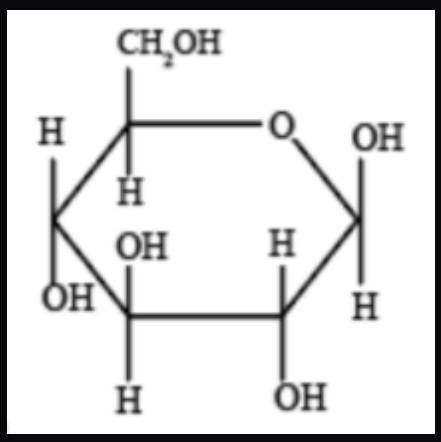
**(A)**



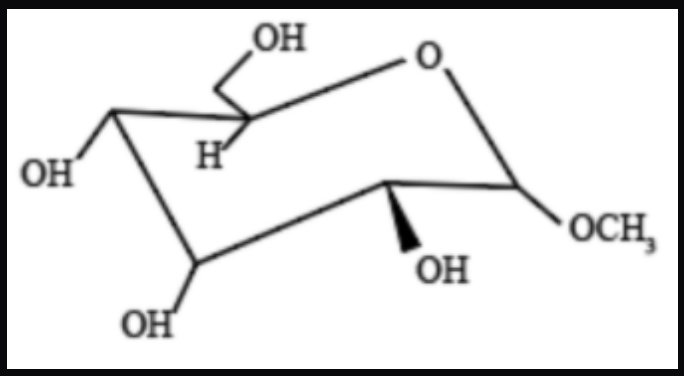
**(B)**




**(C)**




**(D)**




**Q.46**

A 500 g tooth-paste sample has 0.2 g fluoride concentration. What is the concentration of fluoride in terms of ppm level?

**Q.47**

How many litres of water must be added to 1 litre of an aqueous solution of $\mathrm{HCl}$ with a pH of 1 to create an aqueous solution with $\mathrm{pH}$ of $2$?

**Q.48**

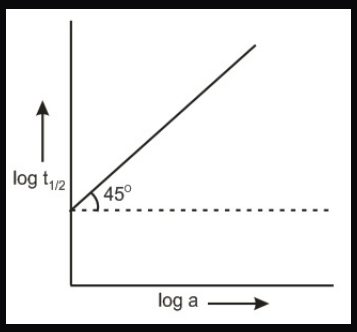
How many of these metals can show $+3$ oxidation state also.
$\mathrm{Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Zn}$

**Q.49**

A weak field complex of $\mathrm{Ni}^{2+}$ has magnetic moment of $2.82 \mathrm{BM}$. The number of electron in the $\mathrm{t}_{2 \mathrm{~g}}$ level of $\mathrm{Ni}^{2+}$ will be

**Q.50**

Following is the graph between $\log \mathrm{t}_{1/2}$ and $\log \mathrm{a}$ (a = initial concentration) for a given reaction at $27^{\circ} \mathrm{C}$. Hence, the order is




**Q.51**

Let $\Delta ABC$ is an isosceles triangle with $AB = AC$. If $B = (0, a),\ C = (2a, 0)$ and the equation of $AB$ is $3x - 4y + 4a = 0$, then the equation of side $AC$ is

**(A)** $y = 8x - 16a$

**(B)** $3y = 4x - 8a$

**(C)** $x = 2a$

**(D)** $y + 8x = 16a$

**Q.52**

The coefficient of $x^2$ in the expansion of $\left(1 - x + 2x^2\right)\left(x + \frac{1}{x}\right)^{10}$ is

**(A)** $210$

**(B)** $714$

**(C)** $504$

**(D)** $240$

**Q.53**

A group of 10 items has arithmetic mean 6. If the arithmetic mean of 4 of these items is $7.5$, then the mean of the remaining items is

**(A)** 6.5

**(B)** 5.5

**(C)** 4.5

**(D)** 5.0

**Q.54**

Let A and B be the sets {1, 2,....10} and {1, 2....,20} respectively. A function is selected randomly from A to B, the probability that the function is non - decreasing is

**(A)** $\frac{^{29}C_{10}}{(20)^{10}}$

**(B)** $\frac{^{29}C_{20}}{(20)^{10}}$

**(C)** $\frac{^{29}C_{11}}{(20)^{10}}$

**(D)** $\frac{^{29}C_{9}}{(20)^{10}}$

**Q.55**

If the vectors
$(x^2-1)\hat{i}+2(x^2-1)\hat{j}-3(x^2-1)\hat{k}$,
$(2x^2-1)\hat{i}+(2x^2+1)\hat{j}+x^2\hat{k}$ and
$(3x^2+2)\hat{i}+(x^2+4)\hat{j}+(x^2+1)\hat{k}$
are non-coplanar, then the number of real values $x$ cannot take is

**(A)** $1$

**(B)** $2$

**(C)** $4$

**(D)** $6$

**Q.56**

If $\int x^3e^{5x}dx=\frac{e^{5x}}{5^4}(f(x))+C$, then $f(x)$ is

**(A)** $\frac{x^3}{5}-\frac{3x^2}{5^2}+\frac{6x}{5^3}-\frac{6}{5^4}$

**(B)** $5x^3-5^2x^2+5^3x-6$

**(C)** $562x^3-15x^2+30x-6$

**(D)** $5^3x^3-75x^2+30x-6$

**Q.57**

If $\overrightarrow{\mathrm{a}}=\hat{\mathrm{i}}-2\hat{\mathrm{j}}+3\hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{b}}=-3\hat{\mathrm{i}}+\hat{\mathrm{j}}-\hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{r}}\times\overrightarrow{\mathrm{a}}=\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{r}}\times\overrightarrow{\mathrm{b}}=\overrightarrow{\mathrm{a}}\times\overrightarrow{\mathrm{b}}$ then a unit vector in the direction of $\overrightarrow{\mathrm{r}}$ is

**(A)** $\frac{1}{3}(-2\hat{\mathrm{i}}+\hat{\mathrm{j}}-\hat{\mathrm{k}})$

**(B)** $\frac{1}{3}(-2\hat{\mathrm{i}}-\hat{\mathrm{j}}+2\hat{\mathrm{k}})$

**(C)** $\frac{1}{3}(-2\hat{i}-\hat{j}-2\hat{k})$

**(D)** None of these

**Q.58**

The domain of the function $f(x) = \exp\left(\sqrt{5x - 3 - 2x^2}\right)$ is

**(A)** $[3/2, \infty)$

**(B)** $[1, 3/2]$

**(C)** $(-\infty, 1]$

**(D)** $(1, 3/2)$

**Q.59**

The number of different garlands, that can be formed using 3 flowers of one kind and 3 flowers of other kind, is

**(A)** $60$

**(B)** $20$

**(C)** $4$

**(D)** $10$

**Q.60**

The value of the determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos(nx) & \cos(n+1)x \\ \sin(\mathrm{n}-1)\mathrm{x} & \sin(nx) & \sin(n+1)x \end{vmatrix}$ is zero, if

**(A)** $\sin x = 0$

**(B)** $\cos x = 0$

**(C)** $a = 0$

**(D)** $\cos x = \frac{1+a^2}{2a}$

**Q.61**

For a complex number $Z$, if the argument of $(Z - a)\left(\overline{Z} - b\right)$ is $\frac{\pi}{4}$ or $\frac{-3\pi}{4}$ (where $a$, $b$ are two real numbers), then the value of $ab$ such that the locus of $Z$ represents a circle with centre $\frac{3}{2} + \frac{i}{2}$ is (where, $i^2 = -1$)

**(A)** $1$

**(B)** $2$

**(C)** $3$

**(D)** $4$

**Q.62**

The number of real values of $k$ for which the lines $\frac{x-k}{4} = \frac{y-1}{2} = \frac{z+1}{1}$ and $\frac{x-(k+1)}{1} = \frac{y}{-1} = \frac{z-1}{2}$ are intersecting is

**(A)** $0$

**(B)** $2$

**(C)** $1$

**(D)** Infinite

**Q.63**

$P = \sqrt{5}\left\{\left(\sqrt{5}+1\right)^{50} - \left(\sqrt{5}-1\right)^{50}\right\}$ is

**(A)** an irrational number

**(B)** an even positive integer

**(C)** an odd positive integer

**(D)** a negative integer

**Q.64**

$\int_{-3}^{2}\{|x+1|+|x+2|+|x-1|\}dx$ is equal to

**(A)** $\frac{31}{2}$

**(B)** $\frac{35}{2}$

**(C)** $\frac{47}{2}$

**(D)** $\frac{39}{2}$

**Q.65**

If $|a|<1$ and $|b|<1$, then the sum of the series $a(a+b)+a^2\left(a^2+b^2\right)+a^3\left(a^3+b^3\right)+\ldots$ is

**(A)** $\frac{a}{1-a}+\frac{ab}{1-ab}$

**(B)** $\frac{a^2}{1-a^2}+\frac{ab}{1-ab}$

**(C)** $\frac{b}{1-b}+\frac{a}{1-a}$

**(D)** $\frac{b^2}{1-b^2}+\frac{ab}{1-ab}$

**Q.66**

Let $f(x)=(x-1)^2(x-2)^3e^x$. Then

**(A)** $\mathrm{f(x)}$ has local maximum at $\mathrm{x}=1$

**(B)** $\mathrm{f(x)}$ has point of inflexion at $\mathrm{x}=1$

**(C)** $\mathrm{f(x)}$ has local minimum at $\mathrm{x}=2$

**(D)** all of these

**Q.67**

If $B, C$ are square matrices of same order such that $C^2 = BC - CB$ and $B^2 = -I$, where $I$ is an identity matrix, then the inverse of matrix $(C - B)$ is

**(A)** $C$

**(B)** $C + B$

**(C)** $C - B$

**(D)** $I$

**Q.68**

If $x$ is real, then the least value of the expression $\frac{x^2-6x+5}{x^2+2x+1}$ is

**(A)** $-1$

**(B)** $-\frac{1}{2}$

**(C)** $-\frac{1}{3}$

**(D)** $1$

**Q.69**

If $\mathrm{y}^2 = \mathrm{P}(\mathrm{x})$ is a polynomial of degree 3 , then $2\frac{d}{dx}\left[y^3 \frac{d^2y}{dx^2}\right]$ is equal to

**(A)** $\mathrm{P}(\mathrm{x}) + \mathrm{P}'(\mathrm{x})$

**(B)** $\mathrm{P}(\mathrm{x}) + \mathrm{P}'''(\mathrm{x})$

**(C)** $\mathrm{P}(\mathrm{x})\mathrm{P}'''(\mathrm{x})$

**(D)** None of these

**Q.70**

If $f : R \to R, f(x) = \frac{\sin([x]\pi)}{x^2+2x+3} + 2x - 1 + \sqrt{x(x-1) + \frac{1}{4}}$ ( where $[x]$ denotes greatest integral value less than or equal to $x$) denotes a function, then number of real solutions of equation $f(x) = f^{-1}(x)$ is

**(A)** $0$

**(B)** $1$

**(C)** $2$

**(D)** $3$

**Q.71**

The number of points at which the function $f(x) = (x - |x|)^2(1 - x + |x|)^2$ is not differentiable in the interval $(-3, 4)$ is

**Q.72**

There are eight rooms on the first floor of a hotel, with four rooms on each side of the corridor, symmetrically situated

(that is each room is exactly opposite to one other room). Four guests have to be accommodated in four of the eight rooms (that is, one in each) such that no two guests are in adjacent rooms or in opposite rooms. If $N$ is the number of ways in which guests can be accommodated. Then the value of $\frac{N}{6}$ is

**Q.73**

If $x\ dy = y\ dx + y^2 dy$ and $y(1) = 1$ then find the value of $y(-3)$

**Q.74**

Let $A = \{1, 2, 3, 4\}$. The number of different unordered pairs $(B, C)$ that can be formed such that $B \subseteq A, C \subseteq A$ and $B \cap C = \phi$, is

**Q.75**

If $\mathrm{A} = \begin{bmatrix} \frac{-1+i\sqrt{3}}{2i} & \frac{-1-i\sqrt{3}}{2i} \\ \frac{1+i\sqrt{3}}{2i} & \frac{1-i\sqrt{3}}{2i} \end{bmatrix}$, $i = \sqrt{-1}$ and $f(x) = x^2 + 2$. Then $\left| \frac{f(\mathrm{A})}{\left(2+i\sqrt{3}\right)} \right|$ is equal to

## Answers & Solutions

**Q.1 Answer:**

$h \geq \frac{5}{2}r$

**Solution:**

**Loss in PE between $A$ and $D$ = gain in KE between $A$ and $D$**

$$mg(h - 2r) = \frac{1}{2}m\left(v^2 - 0\right) \quad (\because K_A = 0)$$

$$v^2 = 2g(h - 2r)$$

**If the block is to complete the loop path then at D**

$$\frac{mv^2}{r} \geq mg$$

$$v^2 \geq rg$$

**From Eqs. (i) and (ii)**

$$2g(h - 2r) \geq rg$$

$$h \geq \frac{5}{2}r$$

**Q.2 Answer:**

$10.5\,\mathrm{RT_0}$

**Solution:**

**Temperature at** (a) $= \mathrm{T_0} = \frac{\mathrm{PV}}{\mathrm{R}}$

**At** (a) $\mathrm{T_0} = \frac{\mathrm{P_0V_0}}{\mathrm{R}}$

**At (c)** $\mathrm{T_C} = \frac{(2\mathrm{P_0})(4\mathrm{V_0})}{\mathrm{R}} = 8\mathrm{T_0}$

$\Delta\mathrm{U} = \mathrm{nC_V}(\mathrm{T_f} - \mathrm{T_i}) = \frac{3}{2}\mathrm{R}(8\mathrm{T_0} - \mathrm{T_0})$

$\Delta U = \frac{21 RT_0}{2} = 10.5 RT_0$

**Q.3 Answer:**

$V_1$

**Solution:**

Average speed of gas molecules is $\sqrt{\frac{8kT}{\pi m}}$. It depends on temperature and molecules mass. So the average speed of $O_2$ will be same in $(A)$ and (C)

**Q.4 Answer:**

$4 : 3$

**Solution:**

Number of emission spectral lines

$N = \frac{n(n-1)}{2}$

$\therefore 3 = \frac{n_1(n_1-1)}{2}$, in first case.

Or $n_1^2 - n_1 - 6 = 0$ or $(n_1 - 3)(n_1 + 2) = 0$

Take positive root.

$\therefore n_1 = 3$

Again, $6 = \frac{n_2(n_2-1)}{2}$, in second case.

Or $n_2^2 - n_2 - 12 = 0$ or $(n_2 - 4)(n_2 + 3) = 0.$

Take positive root, or $n_2 = 4$

Now velocity of electron $v = \frac{2\pi KZe^2}{nh}$

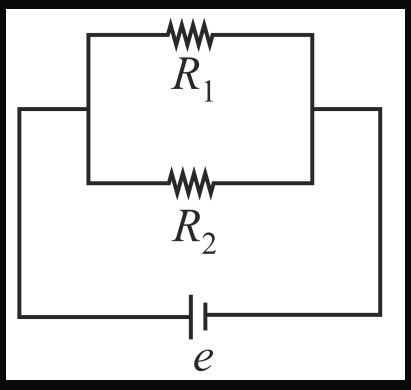
$\therefore \frac{v_1}{v_2} = \frac{n_2}{n_1} = \frac{4}{3}.$

**Q.5 Answer:**

$\frac{B\omega r^2}{2R_1}$

**Solution:**

The equivalent diagram is




The induced emf across the centre and any point on the circumference is

$\varepsilon = \frac{1}{2} B\omega\ell^2 = \frac{B\omega r^2}{2}$

$\therefore$ Current through $R_1 = \frac{B\omega r^2}{2R_1}$

**Q.6 Answer:**

50 N

**Solution:**

$F = v\frac{dm}{dt} = 10 \times 5$ **N = 50 N**

**Q.7 Answer:**

$1 : \sqrt{3}$

**Solution:**

**The equation of motion in exclusive**

$v^2 = 2g\sin 60^\circ x_1 \ \ldots (1)$

$v^2 = 2g\sin 30^\circ x_2$

**Equating,**

$2g\sin 60^\circ x_1 = 2g\sin 30^\circ x_2$

$x_1 : x_2 = 1 : \sqrt{3}$

**Q.8 Answer:**

$\frac{K_1\theta_1 + K_2\theta_2}{K_1 + K_2}$

**Solution:**

**In the given question we have to calculate temperature at junction.Suppose the temperature at junction is** $\theta$**. In steady state the heat current will be same in both rods**

$\therefore H_1 = H_2$

**Thermal conductivity of rods are** $\mathrm{K}_1$ and $\mathrm{K}_2$ **. The temperature at end of both rods are** $\theta_1$ and $\theta_2$**.Length of wire is** $l$ **and area of cross-section is** $\mathrm{A}$**.**

**So we can write**

$\frac{\theta_1 - \theta}{\left(\frac{l}{K_1 A}\right)} = \frac{\theta - \theta_2}{\left(\frac{l}{K_2 A}\right)}$ **(**$\theta$ **=temperature of junction )**

$K_1(\theta_1 - \theta) = K_2(\theta - \theta_2)$

$\theta = \frac{K_1\theta_1 + K_2\theta_2}{K_1 + K_2}$**.**

**Q.9 Answer:**

$0.9 \text{ m}$

**Solution:**

**Here,** $n = 200 \text{ Hz}, v = 360 \text{ m/s}$

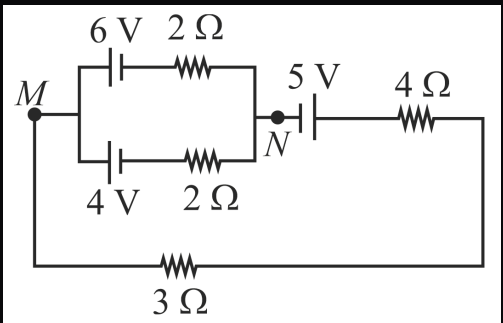
$\therefore \lambda = \frac{v}{n} = \frac{360}{200} = 1.8 \text{ m}$

**Distance between two consecutive anti-nodes** $= \frac{\lambda}{2} = \frac{1.8}{2} = 0.9 \text{ m}$

**Q.10 Answer:**

$(1/2)$ A

**Solution:**



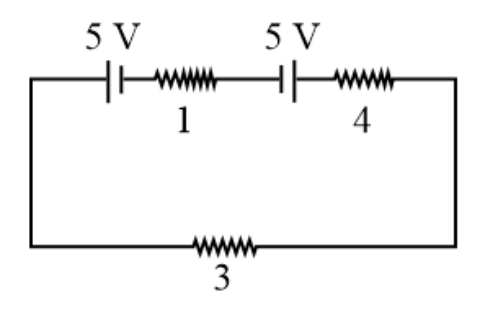

**Equivalent EMF between points $M$ and $N$,**

$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{R_1} + \frac{\varepsilon_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{\frac{6}{2} + \frac{4}{2}}{\frac{1}{2} + \frac{1}{2}}$

$\varepsilon_{eq} = \frac{3+2}{1} = 5 \text{ V}$

**Equivalent resistance between $M$ and $N$,**

$\frac{1}{R} = \frac{1}{2} + \frac{1}{2} \Rightarrow R = 1\ \Omega$




**Equivalent emf of circuit,**

$\varepsilon_{eq} = 5 - 5 = 0$

**So, there will be no current in main circuit.**

**Current will only flow in small loop in the upper circuit.**

**Applying KVL in small loop,**

$6 - 4 - 4i = 0$

**Hence, current through $6$ V battery is $\frac{1}{2}$ A.**

**Q.11 Answer:**

$\frac{\omega^2 R^2}{g}$

**Solution:**

$g_{eff}$ due to rotation $= g_{eff}$ due to depth d

$g - R\omega^2 = g\left(1 - \frac{d}{R}\right) \quad \Rightarrow \quad R\omega^2 = \frac{gd}{R} \Rightarrow d = \frac{R^2\omega^2}{g}$

**Q.12 Answer:**

Speed of light in medium B is three fourth of that in medium A

**Solution:**

$\frac{\sin i}{\sin r} = \cot 37 = \frac{4}{3} = \frac{V_A}{V_B}$

**as going from rarer to denser no TIR can take place**

$\frac{\mu_B}{\mu_A} > 1 \Rightarrow \mu_B > \mu_A$

**Q.13 Answer:**

$h' = h$

**Solution:**

Rise in capillary tube is independent of block inside it.

**Q.14 Answer:**

$-\frac{1}{2}, \frac{1}{2}$ and $\frac{5}{2}$

**Solution:**

$$[E] = [G]^p[h]^q[c]^r$$
$$\left[M^1L^2T^{-2}\right] = \left[M^{-1}L^3T^{-2}\right]^p \left[ML^2T^{-1}\right]^q \left[LT^{-1}\right]^r$$
$$[M]^1[L]^2[T]^{-2} = [M]^{-p+q}[L]^{3p+2q+r}[T]^{-2p-q-r}$$

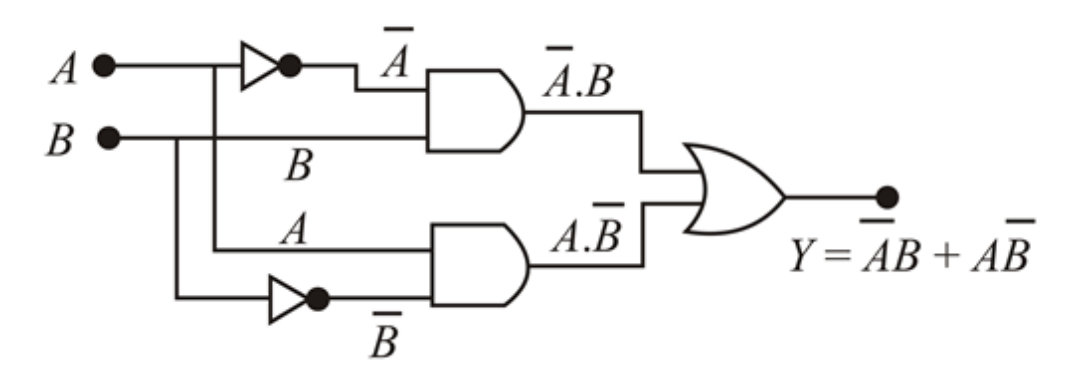
Equating dimensions of M, L and T, we get, $-p + q = 1$,

$$3p + 2q + r = 2$$
$$-2p - q - r = -2$$
$$p = -\frac{1}{2}, q = \frac{1}{2}, r = \frac{5}{2}$$

**Q.15 Answer:**

| $A$ | $B$ | $Y$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**Solution:**




The truth table for the given logic circuit is

Misplaced \hline

**Q.16 Answer:**

40 A and 16 A

**Solution:**

Here, $\eta = 80\%$

$P_i = 4\ \text{kW} = 4000\ \text{W}$

$V_p = 100\ \text{V}, V_s = 200\ \text{V}$

$$I_p = \frac{P_i}{V_p} = \frac{4000}{100} = 40\ \text{A}$$

$$\eta = \frac{V_s I_s}{V_p I_p}; \frac{80}{100} = \frac{200 I_s}{4000} = \frac{I_s}{20}$$

$$\text{or } I_s = \frac{20 \times 80}{100} = 16\ \text{A}$$

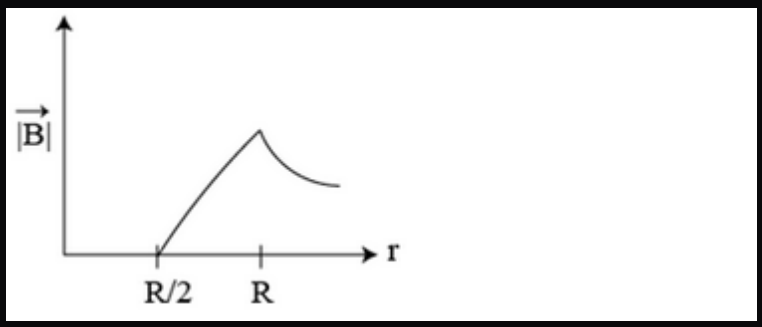
**Q.17 Answer:**

Phase of the oscillator is same at $\text{t} = 1$ s and $\text{t} = 5$ s.

**Solution:**

**The phase of a particle executing SHM is defined as the state of a particle as regard to its position and direction of motion at any instant of time. In the given curve, phase is same when $\text{t} = 1$ s and $\text{t} = 5$ s. Also phase is same when $\text{t} = 2$ s and $\text{t} = 6$ s .**

**Q.18 Answer:**

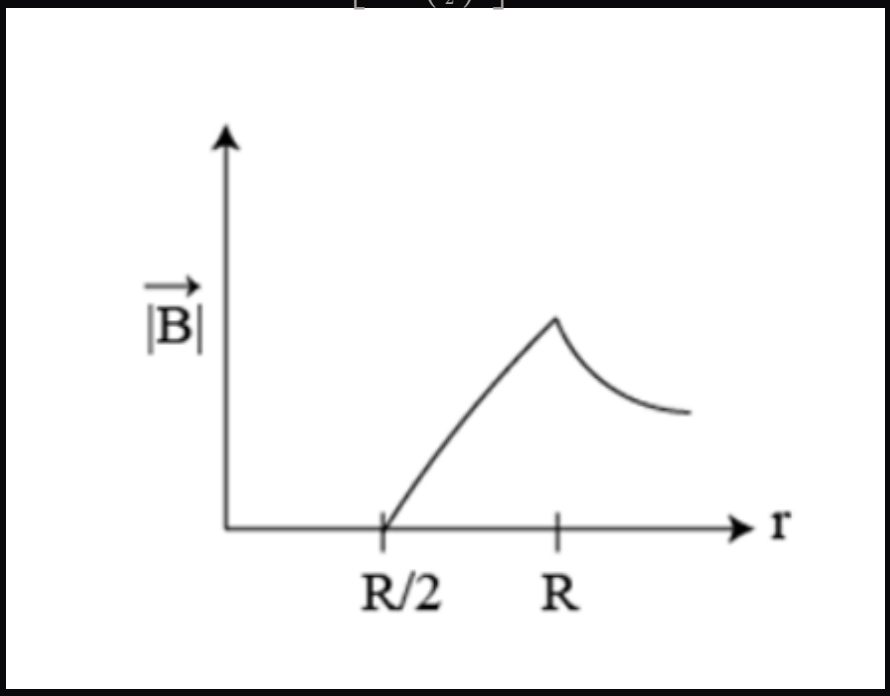



**Solution:**

$r < \frac{R}{2}, B = 0$

$r > R, B = \frac{\mu_0 I}{2\pi r}$

$\frac{R}{2} < r < R, B = \frac{\mu_0 I}{2\pi r}\left[\frac{r^2 - \left(\frac{R}{2}\right)^2}{R^2 - \left(\frac{R}{2}\right)^2}\right]$




**Q.19 Answer:**

$\frac{4\sqrt{2}Gm^2}{r}$

**Solution:**

$W_{\text{ext}} = \Delta V \times m = [V_f - V_0]\, m$
$= \left[0 - 4\left(-\frac{Gm}{r/\sqrt{2}}\right)\right] \times m$
$W_{\text{ext}} = \frac{4\sqrt{2}Gm^2}{r}$

**Q.20 Answer:**

$E^{\frac{1}{2}}$

**Solution:**

**Wavelength of the photon interms of energy will be given by,**

$\lambda_2 = \frac{hc}{E}$

**De-Broglie of the proton will be,**

$\lambda_1 = \frac{h}{p} = \frac{h}{\sqrt{2m_pE}}$

**(where $p$ is the momentum of the proton and $m_p$ is the mass of the proton)**

**From above two equations,**

$\frac{\lambda_1}{\lambda_2} = \frac{\frac{\mathrm{h}}{\sqrt{2\mathrm{m_pE}}}}{\frac{\mathrm{hc}}{\mathrm{E}}} = \frac{\sqrt{\mathrm{E}}}{\mathrm{c}\sqrt{2\mathrm{m_p}}}$

$\Rightarrow \frac{\lambda_1}{\lambda_2} \propto E^{\frac{1}{2}}$

**Q.21 Answer:**

—

**Solution:**

Let ground state energy (in eV) be $E$ -

Then, from the given condition

$E_{2n} - E_1 = 204\text{eV}$ Or $\frac{E_1}{4n^2} - E_1 = 204\text{eV}$ Or $\mathrm{E}_1\left(\frac{1}{4\mathrm{n}^2} - 1\right) = 204\text{eV}$ . . . (i) And $E_{2n} - E_n = 40.8\text{eV}$ Or $\frac{E_1}{4n^2} - \frac{E_1}{n^2} = 40.8\text{eV}$ or $\mathrm{E}_1 = \left(\frac{-3}{4\mathrm{n}^2}\right) = 40.8\text{eV}$ . . . . (ii)

Or $\frac{E_1}{4n^2} - \frac{E_1}{n^2} = 40.8\text{eV}$

or $\mathrm{E}_1 = \left(\frac{-3}{4\mathrm{n}^2}\right) = 40.8\text{eV}$ . . .. (ii)

From Eqs. (i) and (ii)

$\frac{1-\frac{1}{4n}}{\frac{3}{4n^2}} = 5$

Or $1 = \frac{1}{4n^2} + \frac{15}{4n^2}$

Or $\frac{4}{n^2} = 1$

or $n = 2$

From Eq. (ii),

$\mathrm{E}_1 = -\frac{4}{3}\mathrm{n}^2(40.8)\text{eV}$

$= -\frac{4}{3}(2)^2(40.8)eV$

or $\mathrm{E}_1 = -217.6\text{eV}$

$E_1 = -(13.6)Z^2$

Therefore $Z^2 = \frac{E_1}{-13{\cdot}6} = \frac{-217{\cdot}6}{-13{\cdot}6} = 16$

Therefore $Z = 4$

**Q.22 Answer:**

—

**Solution:**

It is given that two trains are of equal length, $L_1 = L_2 = 120\ \text{m}$. The first train takes time, $t_1 = 10\ \text{s}$ to cross a telephone pole, while the second takes $t_2 = 15\ \text{s}$

Crossing a telephone requires a train to travel a distance equal to its own length.

Speed of the first train $v_1 = \frac{L_1}{t_1} = \left(\frac{120}{10}\right)\frac{\text{m}}{\text{sec}} = 12\ \text{m}\ \text{s}^{-1}$.

Speed of the second train $v_2 = \left(\frac{120}{15}\right)\frac{\text{m}}{\text{sec}} = 8\ \text{m}\ \text{s}^{-1}$.

Relative speed when travelling in opposite direction,$v_{rel} = (v_1 + v_2) = \left(12 + 8\right) = 20\ \text{m}\ \text{s}^{-1}$

Hence, time required for the trains to cross each other completely, $= \frac{(120+120)}{20}\ \text{s} = 12\ \text{s}$.

**Q.23 Answer:**

—

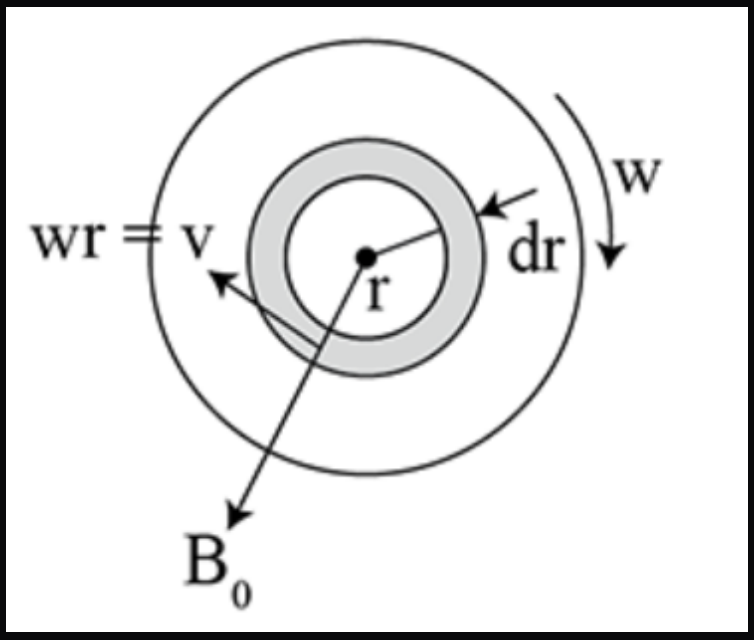
**Solution:**

The force an any small part of the disc is in the vertically upward direction

$$dF = \left(\frac{Q}{\pi R^2} 2\pi r dr\right) \omega r B_0$$

$$dF = \frac{2Q\omega B_0}{R^2} r^2 dr$$

$$F = \frac{2}{3} Q\omega B_0 R = Mg$$




$\Rightarrow \omega = \frac{3Mg}{2QB_0R} = \frac{3\times2\times10}{2\times2\times10^{-2}\times10\times\frac{1}{6}} = 9\times10^2 \text{rads}^{-1}$

**Q.24 Answer:**

—

**Solution:**

For refraction at the first surface, $u = -8$ cm,

$R_1 = -8$ cm, $\mu_1 = 1, \mu_2 = 1.5$

$\Rightarrow \quad \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1}$

$\Rightarrow \quad \frac{1.5}{v'} + \frac{1}{8} = \frac{0.5}{-8} \Rightarrow v' = -8$ cm

It means due to the first surface the image is formed at the centre.

For the second surface $u = -9$ cm, $\mu_1 = 1.5, R_2 = -9$ cm

$\Rightarrow \quad \frac{m_2}{v} - \frac{m_1}{u} = \frac{m_2 - m_1}{R_2}$

$\Rightarrow \quad \frac{1}{v} + \frac{1.9}{5} = \frac{1-1.5}{-9} \Rightarrow v = -9$ cm

Thus, the final image is formed at the centre of the sphere.

**Q.25 Answer:**

—

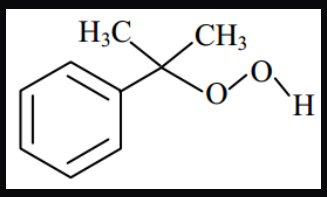
**Solution:**

$$\frac{\Delta p}{p} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$
$$= \frac{0.05}{5} \times 100 + \frac{0.05}{1} \times 100$$

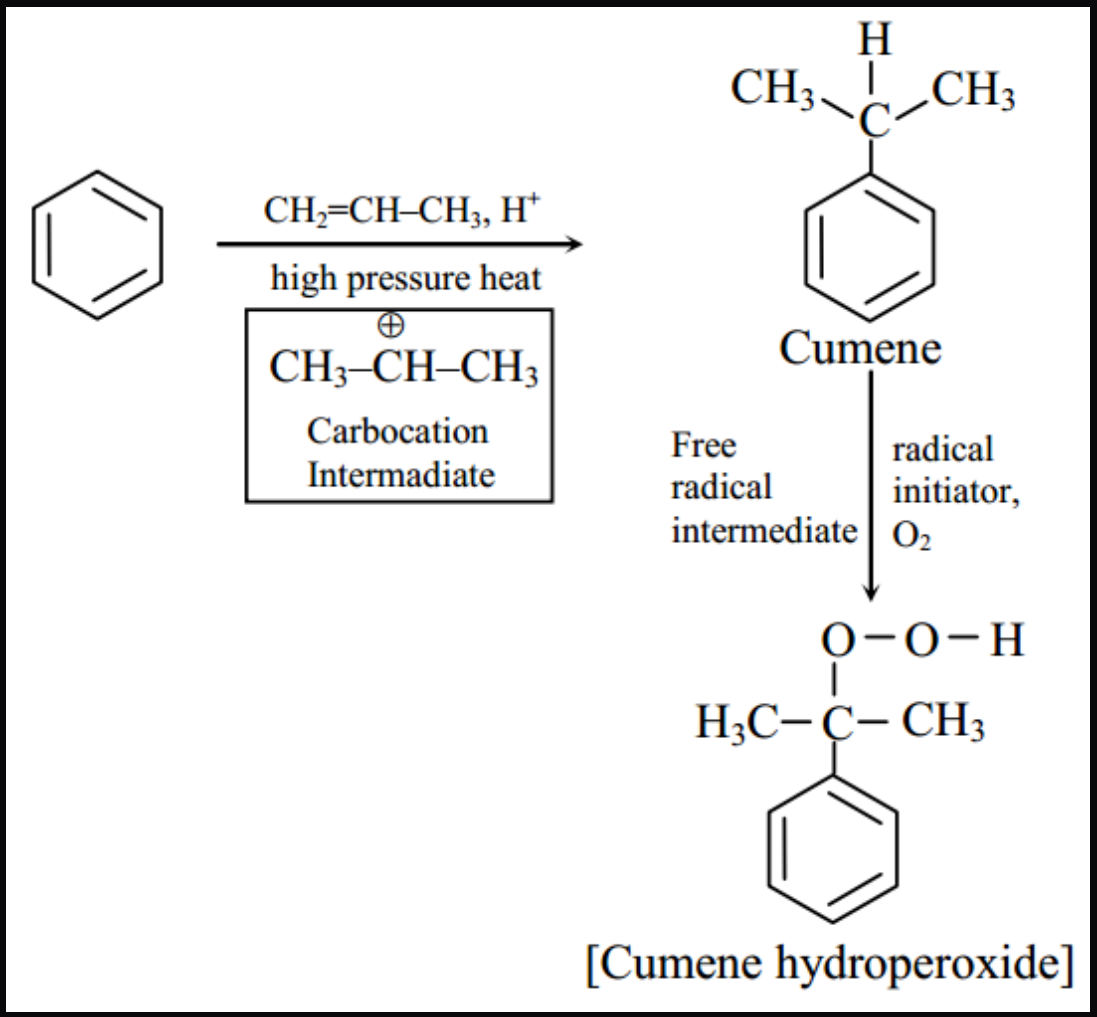
$= 6$

$\frac{\Delta p}{p} = 6\%$

**Q.26 Answer:**



**Solution:**



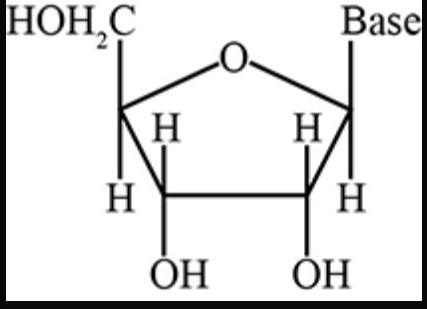

**Q.27 Answer:**

an aldopentose and a nitrogenous base

**Solution:**

**A nucleoside on hydrolysis gives an aldopentose and a nitrogenous base.**

**Nucleoside is a unit formed by the attachment of a base to an aldopentose sugar.**




**Q.28 Answer:**

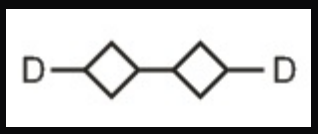
Halogen

**Solution:**

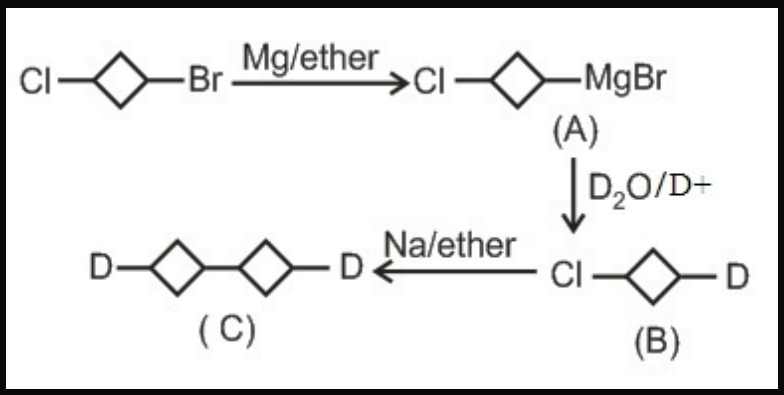
**In a period on moving left to right $Z_{effective}$ increases, so atomic size decreases.**

**For inert gases, Van der waal radius is considered, so it is bigger in size as compared to halogens.**

**Q.29 Answer:**




**Solution:**




**Q.30 Answer:**

$CO_3^{2-}(aq)$

**Solution:**

**On reversing the given reaction**

$CO_3^{2-} + H_2PO_4^- \rightleftharpoons HPO_4^{2-} + HCO_3^{2-} Kc = 10^3$

**Since the value of $K_c$ for this reaction is greater than 1, it means this reaction is favourable in forward direction. Therefore, $CO_3^{2-}$ is the stronger conjugate base than $HPO_4^{2-}$**

**Q.31 Answer:**

$KMnO_4$

**Solution:**

$KMnO_4$ has +7 oxidation state of Mn, which is maximum oxidation state.

**Q.32 Answer:**

$He_2^+ < O_2^- < C_2 < NO$

**Solution:**

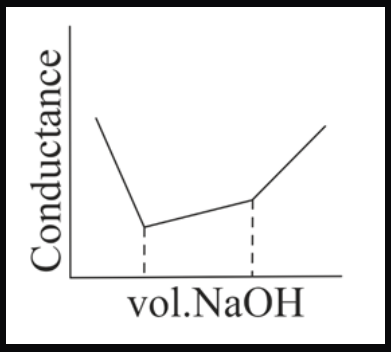
Bond order of $He_2^+ \rightarrow 0.5$

$O_2^- \rightarrow 1.5$

$C_2 \rightarrow 2$

$NO \rightarrow 2.5$

**Q.33 Answer:**




**Solution:**

Molar conductance of $H^+$ and $OH^-$ are very high as compared to other ions

Initially conductance of solutions sharply decreases due to consumption of free $H^+$ ions and then increase due to formation of salt $(NaCN)$ and after complete neutralization further sharply increases due to presence of $OH^-$.

**Q.34 Answer:**

750 K

**Solution:**

$\frac{1}{2}X_2 + \frac{3}{2}Y_2 \longrightarrow XY_3$

$\Delta S^\circ_{reaction} = \Delta S^\circ_{products} - \Delta S^\circ_{reactants}$

$\therefore \Delta S^\circ_{reaction} = \Delta S^\circ_{XY_3} - \frac{1}{2}\Delta S^\circ_{X_2} - \frac{3}{2}\Delta S^\circ_{Y_2}$

$= 50 - \frac{1}{2} \times 60 - \frac{3}{2} \times 40 = -40\ JK^{-1}$

Using equation, $\Delta G = \Delta H - T\Delta S$

We have, $\Delta H = -30\ kJ$, $\Delta S = -40\ JK^{-1}\ mol^{-1}$

and at equilibrium, $\Delta G = 0$. Therefore,

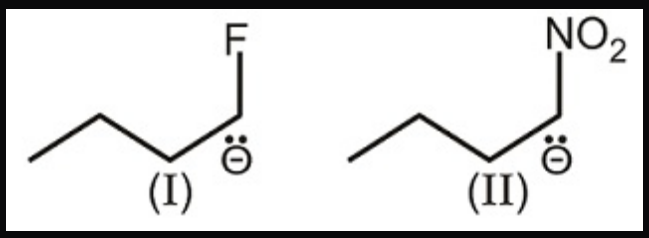
$T = \frac{\Delta H}{\Delta S} = \frac{-30 \times 1000}{-40} = 750\ K$

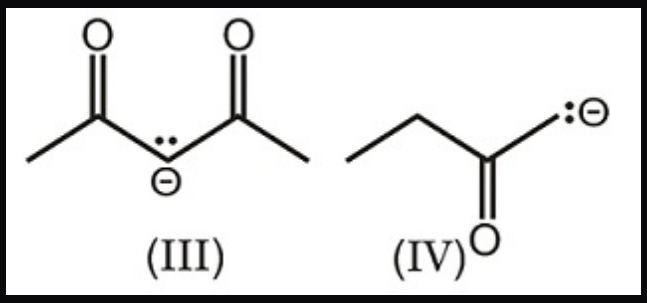
**Q.35 Answer:**

I < IV < II < III

**Solution:**

Carbanions formed after decarboxylation (loss of $CO_2$) are

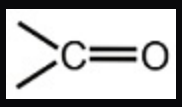



In (I) F is electron withdrawing (-I) effect but no resonance.

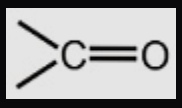
In (II) $-NO_2$ is having (-I) effect and negative charge resonated with $-NO_2$

In (IV) negative charge of carbanion resonates with carbonyl but



is having lower (-I) effect than $-NO_2$ IV is less stable than II

In (III) the negative charge of carbanion resonates with two



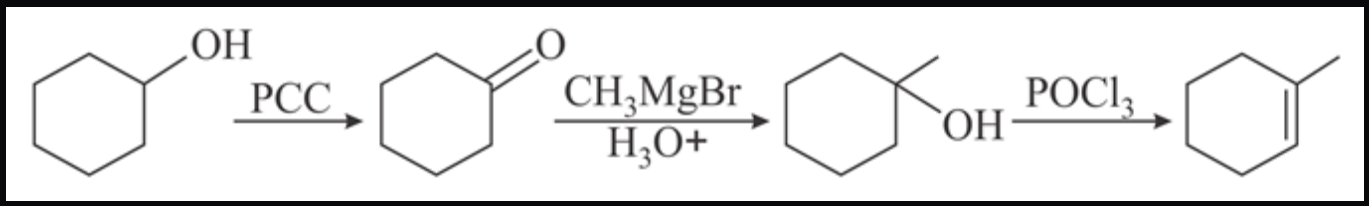
(carbonyl groups) on both sides i.e. most stable

I < IV < II < III is the order of stabilities of carbanions & same is order of reactivities towards thermal decarboxylations.

**Q.36 Answer:**

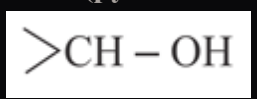
(i) PCC, (ii) MeMgBr, (iii) $POCl_3$

**Solution:**

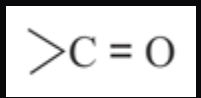



**Explanation**

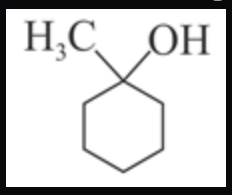
PCC (pyridinuim chloro chromate ) oxidises a secondary hydroxyl group



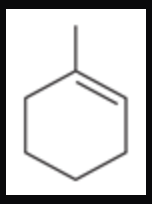
to



**which is nucleophilically added across by $CH_3^- \overset{+}{Mg}Br$ to give**



**undergoes dehydration with $POCl_3$ to give**



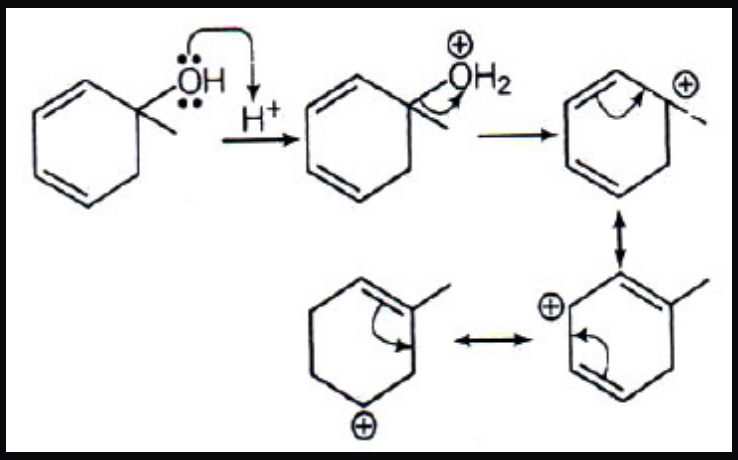
**Q.37 Answer:**

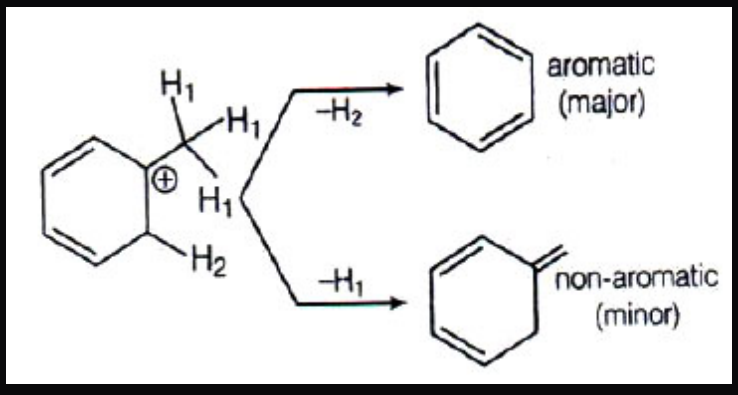
Both statement I and statement II are true and the statement II is the correct explanation of the statement I

**Solution:**

**This problem includes conceptual mixing of conjugation and case of dehydration.**

**Conjugation : Conjugation increases stability of carbocation by increase in number of resonance structures.**




**The correct reason is major product is benzene due to formation of aromatic compound.**

**Q.38 Answer:**

$10^{-1}$

**Solution:**

**Given :** $E_a = 2.303\,RT\ J\ mol^{-1}$

**Temperature**$= T\ K$

**Solution : According to the Arrhenius equation,**

$\ln\ k = \ln\ A - \frac{E_a}{RT}$

**Substituting the given values,**

$\ln\ k = \ln\ A - \frac{2.303\,RT}{RT} \ln\ \frac{k}{A} = -2.303 \log\ \frac{k}{A} = -1 \therefore\ \frac{k}{A} = \text{antilog}\ (-1) \quad \frac{k}{A} = 10^{-1}$

**Q.39 Answer:**

0.5

**Solution:**

**Given:**

$x_A = 0.2\ \text{So},\ \ x_B = 0.8$

**The formula for total pressure:**

$P_T = X_A P_A^\circ + X_B P_B^\circ \quad = 0.2 \times 60 + 0.8 \times 15 \quad = 24\ \text{torr}$

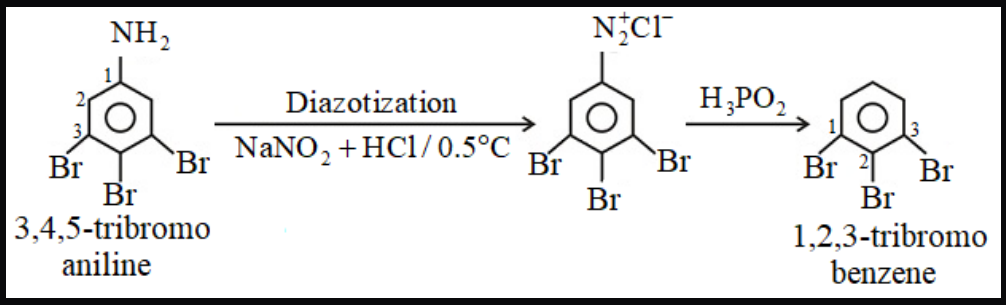
**The Mole fraction of A in vapour phase:**

$y_A = \frac{P_A}{P_T} = \frac{12}{24} = 0.5$

**Q.40 Answer:**

1,2,3 - Tribromobenzene

**Solution:**

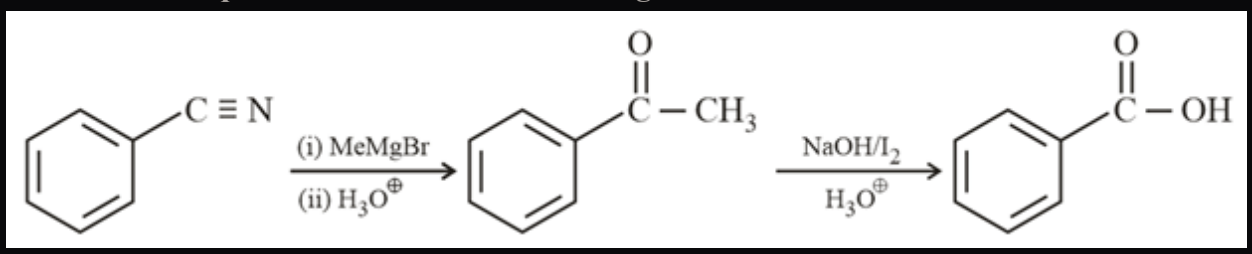



**Q.41 Answer:**

iii

**Solution:**

**The reaction sequence of the above reaction is given below:**




**In the first step, nucleophilic addition reaction occurs at the nitrile group and after hydrolysis, acetophenone is formed. In the second step, haloform reaction occurs and it is oxidised to benzoic acid.**

**Q.42 Answer:**

i, ii, iii

**Solution:**

(i) $PH_5$ and $BiCl_5$ do not exist because electronegativity of H and Cl is not so high as F and O. and size of Cl and H are large.
(ii) $p\pi - d\pi$ bond is present in $SO_2$( s has d orbital).
(iii) $I_3^+$ has bent geometry( 2 bond pairs and 2 lone pairs).
(iv) $SeF_4$ is of see-saw shape.

**Q.43 Answer:**

3, 3, 3

**Solution:**

The complex can be written as follows.
$A - [Co(NH_3)_6]Cl_3$
$B - [Co(NH_3)_5 Cl]Cl_2$
$C - [Co(NH_3)_4 Cl_4]Cl$
Primary Valancy of a Metal atom is equal to its Oxidation state, and in all the given compunds, Co is present in +3 oxidation state.

**Q.44 Answer:**

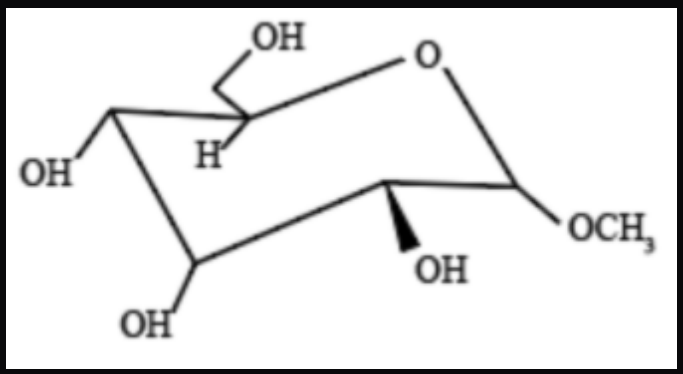
conc. $H_2SO_4$ /170°C

**Solution:**

When alcohols are heated with a protonic acid such as conc.$H_2SO_4$ or $H_3PO_4$ at 170°C, they get dehydrated to form alkenes. The following reaction will take place when ethanol is treated with conc. sulphuric acid to form ethylene-

$$CH_3CH_2OH \xrightarrow{H_2SO_4,\ 170^\circ C} CH_2 = CH_2 + H_2O$$

**Q.45 Answer:**




**Solution:**

As compound having free anomeric carbon is reducing in nature. Therefore D is non-reducing as in it there is no free anomeric carbon.

**Q.46 Answer:**

—

**Solution:**

**As 500 g of toothpaste** $\equiv 0.2$ **g of** $\mathrm{F}^{-}$
**So** $10^{6}$ **g of toothpaste** $\equiv \frac{0.2 \mathrm{~g} \times 10^{6} \mathrm{~g}}{500 \mathrm{~g}} \mathrm{F}^{-}$
$= 400 \mathrm{~g~F}^{-}$
**Hence, concentration in ppm is 400.**

**Q.47 Answer:**

—

**Solution:**

$0.1 \times 1 = (1 + \mathrm{v}) \times 0.01 \Rightarrow 1 + \mathrm{v} = \frac{0.1}{0.01}$
$\Rightarrow 1 + v = 10 \Rightarrow v = 10 - 1 = 9L$

**Q.48 Answer:**

—

**Solution:**

$\mathrm{Sc, Ti, V, Cr, Mn, Fe, Co, Ni}$ **can show** $+3$ **oxidation state also.**

**Q.49 Answer:**

—

**Solution:**

$\mathrm{Ni}^{2+} = (\mathrm{Ar})3 \mathrm{~d}^{8}$
**and as** $\mu = 2.82\mathrm{BM}$
**So** $n = 2$ **Thus,** $\mathrm{t}_{2 \mathrm{~g}}^{6} \mathrm{e}_{\mathrm{g}}^{2}$ **configuration is present here. Hence number of electrons in** $\mathrm{t}_{2 \mathrm{~g}}$ **are** $6$.

**Q.50 Answer:**

—

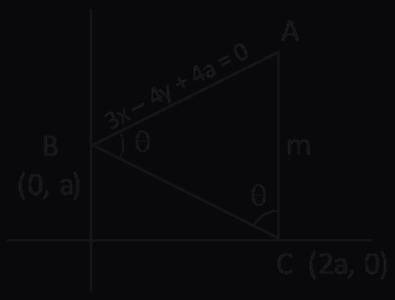
**Solution:**

$\mathrm{A} \rightarrow$ Products
**From graph** $\rightarrow$
$\log \mathrm{t}_{\frac{1}{2}} \propto \tan 45 (= \log \mathrm{a})$
$\log \mathrm{t}_{\frac{1}{2}} \propto \log \mathrm{a}$
$\mathrm{t}_{\frac{1}{2}} \propto \mathrm{a}$ **(Initial Condition)**
**For zero-order reaction**
$\therefore \mathrm{t}_{\frac{1}{2}} \propto (\mathrm{a})^{1-\mathrm{n}}$

**Q.51 Answer:**

$x = 2a$

**Solution:**




**Let,** $\angle ABC = \angle ACB = \theta$ **and slope of** $AC = m$

**Slope of** $BC = -\frac{1}{2}$

**Slope of** $AB = \frac{3}{4}$

**Now, tan** $(\angle ABC) = \tan(\angle \mathrm{ACB})$

$\Rightarrow \frac{\frac{3}{4}-\left(-\frac{1}{2}\right)}{1+\left(\frac{3}{4}\right)\left(-\frac{1}{2}\right)} = \frac{-\frac{1}{2}-m}{1+\left(-\frac{1}{2}m\right)}$

$\Rightarrow 2 = \frac{-\frac{1}{2}-m}{1-\frac{m}{2}} \Rightarrow 2 - m = -\frac{1}{2} - m \Rightarrow m =$ **not defined**

$\Rightarrow$ **equation of** $AC$ **is** $x = 2a$

**Q.52 Answer:**

714

**Solution:**

**In denominator we have** $x^{10}$ **so effectively we need** $x^{12}$ **from the numerator to get overall coefficient of** $x^2$**.**

**Be careful in this step**

**The required coefficient is the coefficient of** $x^{12}$ **in**

$\left(1 - x + 2x^2\right)\left(x^2 + 1\right)^{10}$

$= \left(1 - x + 2x^2\right)\left(1 + {}^{10}C_1 x^2 + {}^{10}C_2 x^4 + \ldots\ldots + {}^{10}C_9 x^{18} + x^{20}\right)$

**Coefficient of** $x^{12}$ **is** $1 \times {}^{10}C_6 + 2 \times {}^{10}C_5 = \frac{10\times 9\times 8\times 7}{4\times 3\times 2} + \frac{2\times 10\times 9\times 8\times 7\times 6}{5\times 4\times 3\times 2}$

$= 210 + 504$

$= 714$

**Q.53 Answer:**

5.0

**Solution:**

**Given that,** $n_1 = 4,\ \overline{x}_1 = 7.5,\ n_1 + n_2 = 10,\ \overline{x} = 6$

$\therefore \quad 6 = \frac{4\times 7.5 + 6\times \overline{x}_2}{10}$

$\Rightarrow \quad 60 = 30 + 6\overline{x}_2$

$\Rightarrow \quad \overline{x}_2 = \frac{30}{6} = 5$

**Q.54 Answer:**

$\frac{{}^{29}C_{10}}{(20)^{10}}$

**Solution:**

Total number of functions from A to B = $(20)^{10}$. The number of non - decreasing function is the number of non - negative integral solutions of the equation

$x_1 + x_2 + \ldots . + x_{10} = 20$ i.e., ${}^{29}C_{10}$

$\Rightarrow$ The required probability $= \frac{{}^{29}C_{19}}{(20)^{10}}$

**Q.55 Answer:**

2

**Solution:**

For non-coplanar vectors

$$\begin{vmatrix} (x^2-1) & 2(x^2-1) & -3(x^2-1) \\ (2x^2-1) & (2x^2+1) & x^2 \\ (3x^2+2) & (x^2+4) & x^2+1 \end{vmatrix} \neq 0$$

$$(x^2-1)\begin{vmatrix} 1 & 2 & -3 \\ 2x^2-1 & 2x^2+1 & x^2 \\ 3x^2+2 & x^2+4 & x^2+1 \end{vmatrix} \neq 0$$

$$(x^2-1)\left\{\left[1\left(2x^4+3x^2+1\right)-x^4-4x^2\right]-2\left(2x^4+x^2-1-3x^4-2x^2\right)-3\left[2x^4+7x^2-4-6x^4-7x^2-2\right]\right\}$$

$\neq 0$

$$(x^2-1)\left[x^4-x^2+1+2x^4+2x^2+2+12x^4+18\right] \neq 0$$

$$(x^2-1)\left[15x^4+x^2+21\right] \neq 0$$

$\Rightarrow x \neq +1, -1$

**Q.56 Answer:**

$5^3x^3 - 75x^2 + 30x - 6$

**Solution:**

We have,

$I = \int x^3 e^{5x} dx = \frac{e^{5x}}{5^4}(f(x)) + C$

Using integration by parts,

$I = e^{5x}\left(\frac{x^3}{5} - \frac{3x^2}{5^2} + \frac{6x}{5^3} - \frac{6}{5^4}\right) + C$

$\Rightarrow I = \frac{e^{5x}}{5^4}\left(5^3x^3 - 75x^2 + 30x - 6\right) + C$

**Q.57 Answer:**

$\frac{1}{3}(-2\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\hat{\mathrm{k}})$

**Solution:**

We have

$$\vec{r} \times \vec{a} + \vec{r} \times \vec{b} = \vec{a} \times \vec{b} + \vec{b} \times \vec{a} = \vec{0}$$

$$\Rightarrow \vec{r} \times (\vec{a} + \vec{b}) = \vec{0}$$

$$\Rightarrow \vec{r} = \lambda(\vec{a} + \vec{b}) = \lambda(-2\hat{i} - \hat{j} + 2\hat{k})$$

$$\Rightarrow \hat{r} = \frac{1}{3}(-2\hat{i} - \hat{j} + 2\hat{k})$$

**Q.58 Answer:**

$[1, 3/2]$

**Solution:**

We have, $f(x) = \exp\left(\sqrt{5x - 3 - 2x^2}\right)$

i.e. $f(x) = e^{\sqrt{5x-3-2x^2}}$

For Domain of $f(x)$, $\sqrt{5x - 3 - 2x^2}$ should be +ve.

i.e., $\sqrt{5x - 3 - 2x^2} \geq 0$

$\Rightarrow 2x^2 - 5x + 3 \leq 0$

(By taking $-$ve sign common)

$\Rightarrow 2x(x - 1) - 3(x - 1) \leq 0$

$\Rightarrow (2x - 3)(x - 1) \leq 0$

$\Rightarrow 2x - 3 \leq 0 \quad$ or $\quad x - 1 \geq 0$

$\Rightarrow x \leq \frac{3}{2} \quad$ or $\quad x \geq 1$

$\therefore 1 \leq x \leq \frac{3}{2} \quad$ i.e., $x \in \left[1, \frac{3}{2}\right]$

Hence, domain of the given function is $\left[1, \frac{3}{2}\right]$.

**Q.59 Answer:**

10

**Solution:**

Let the flowers be of type $A$ and type $B$.

If we arrange the flowers of type $A$ in a circle, there is only one way to do that since they are all identical.

Now we have 3 gaps between the type $A$ flowers where 3 type $B$ flowers must go.

Total number of ways of that arrangement $= {}^{3+3-1}C_{3-1} = {}^{5}C_{2} = 10$

**Q.60 Answer:**

$\sin x = 0$

**Solution:**

**Given determinant**

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos nx & \cos(n+1)x \\ \sin(n-1)x & \sin nx & \sin(n+1)x \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1+a^2-2a\cos x & a & a^2 \\ 0 & \cos nx & \cos(n+1)x \\ 0 & \sin nx & \sin(n+1)x \end{vmatrix} = 0$$

**By applying** $C_1 \to C_1 + C_3 - 2\cos \times C_2$

**By expanding**

$$\left(1+a^2-2a\cos x\right)[\cos nx \sin(n+1)x$$

$$-\sin nx \cos(n+1)x] = 0$$

$$\text{Now }, \left(1+a^2-2a\cos x\right)\sin(n+1-n)x = 0$$

$$\Rightarrow \left(1+a^2-2a\cos x\right)\sin x = 0$$

$$\sin x = 0 \text{ or } \cos x = \frac{1+a^2}{2a}$$

$$\text{As } a \neq 1 \quad \therefore \left(\frac{1+a^2}{2a}\right) > 1$$

$$\Rightarrow \cos x > 1 \text{ It is not possible.}$$

$$\therefore \sin x = 0$$

**Q.61 Answer:**

**2**

**Solution:**

**Let,** $Z = x + iy$**, then**

$$(Z-a)\left(\overline{Z}-b\right) = Z\overline{Z} - a\overline{Z} - bZ + ab$$

$$= \left(x^2+y^2\right) - a(x-iy) - b(x+iy) + ab$$

$$= \left(x^2+y^2\right) - (a+b)x + i(a-b)y + ab$$

**If the argument of the above complex number is** $\frac{\pi}{4}$ **or** $-\frac{3\pi}{4}$**, then**

$$x^2+y^2-(a+b)x+ab = (a-b)y$$

$\Rightarrow$ **Centre of the circle is** $\left(\frac{a+b}{2}, \frac{a-b}{2}\right)$

$$\Rightarrow \frac{a+b}{2} = \frac{3}{2} \ \& \ \frac{a-b}{2} = \frac{1}{2}$$

$$\Rightarrow a = 2,\ b = 1 \Rightarrow ab = 2$$

**Q.62 Answer:**

Infinite

**Solution:**

**Any point on the first line is** $(4r+k\ ,\ 2\ r+1, r-1)$**, and any point on the second line is** $(r'+k+1\ ,\ -r', 2r'+1)$**for some values of** $r$ **and** $r'$**. The lines are intersecting if these two points coincide i.e**

$4r+k = r'+k+1, 2r+1 = -r', \mathrm{r}\ -1 = 2r'+1$ **for some** $r$ **and** $r'$

$\Rightarrow \quad 4r - r' = 1$ , $2\ r + r' = -1, r - 2r' = 2$

**Now,** $4r - r' = 1$ , $2\ r + r' = -1 \ \Rightarrow \ r = 0, r' = -1$ **which satisfy** $r$ -2 $r' = 2$.

$\Rightarrow$ **The given lines are intersecting for all real values of** $k$.

**Q.63 Answer:**

an even positive integer

**Solution:**

**We have,** $P = \sqrt{5}\left\{\left(\sqrt{5}+1\right)^{50} - \left(\sqrt{5}-1\right)^{50}\right\}$

$= 10\left[{}^{50}C_1 5^{24} + {}^{50}C_3 5^{23} + \ldots + {}^{50}C_{49}\right]$

**(which, is an even positive integer).**

**Q.64 Answer:**

$\frac{47}{2}$

**Solution:**

**Let** $I = \int_{-3}^{2}(|x+1| + |x+2| + |x-1|\ )dx$

**Again, let** $f(x) = |x+1| + |x+2| + |x-1|$

$$= \begin{cases} -(x+1) - (x+2) - (x-1), & -3 < x \leq -2 \\ -(x+1) + x + 2 - (x-1), & -2 < x \leq -1 \\ 1 + x + x + 2 - (x-1) & -1 < x \leq 0 \\ 1 + x + x + 2 - (x-1) & 0 \leq x < 1 \\ 1 + x + x + 2 + x - 1 & 1 \leq x < 2 \end{cases}$$

$$= \begin{cases} -3x - 2, & -3 < x \leq -2 \\ -x + 2, & -2 < x \leq -1 \\ x + 4, & -1 \leq x < 1 \\ 3x + 2, & 1 \leq x < 2 \end{cases}$$

$\therefore I = \int_{-3}^{-2}(-3x-2)dx + \int_{-2}^{-1}(-x+2)dx$

$+ \int_{-1}^{1}(x+4)dx + \int_{1}^{2}(3x+2)dx$

$= \left[-\frac{3x^2}{2} - 2x\right]_{-3}^{-2} + \left[-\frac{x^2}{2} + 2x\right]_{-2}^{-1}$

$+ \left[\frac{x^2}{2} + 4x\right]_{-1}^{1} + \left[\frac{3x^2}{2} + 2x\right]_{1}^{2}$

$= \left[-6 + 4 - \left(-\frac{27}{2} + 6\right)\right] + \left[-\frac{1}{2} - 2 - \left(-2 - 4\right)\right]$

$+ \left[\frac{1}{2} + 4 - \left(\frac{1}{2} - 4\right)\right]\left[6 + 4 - \left(\frac{3}{2} + 2\right)\right]$

$= \frac{11}{2} + \frac{7}{2} + 8 + \frac{13}{2}$

$= \frac{31}{2} + 8 = \frac{47}{2}$

**Alternate**

**Let** $I = \int_{-3}^{2}\{|x+1| + |x+2| + |x-1|\}dx$

$= \int_{-3}^{-1}|x+1|dx + \int_{-1}^{2}|x+1|dx + \int_{-3}^{-2}|x+2|dx$

$+ \int_{-2}^{2}|x+2|dx + \int_{-3}^{1}|x-1|dx$

$+ \int_{1}^{2}|x-1|dx$

$= -\int_{-3}^{-1}(x+1)dx + \int_{-1}^{2}(x+1)dx - \int_{-3}^{-2}(x+2)dx$

$+ \int_{-2}^{2}(x+2)dx - \int_{-3}^{1}(x-1)dx + \int_{1}^{2}(x-1)dx$

$= -\left(\frac{x^2}{2}+x\right)_{-3}^{-1} + \left(\frac{x^2}{2}+x\right)_{-1}^{2} - \left(\frac{x^2}{2}+2x\right)_{-3}^{-2}$
$+\left(\frac{x^2}{2}+2x\right)_{-2}^{2} - \left(\frac{x^2}{2}-x\right)_{-3}^{1} - \left(\frac{x^2}{2}-x\right)_{1}^{2}$
$= \frac{47}{2}$

**Q.65 Answer:**

$\frac{a^2}{1-a^2} + \frac{ab}{1-ab}$

**Solution:**

**We have,** $|a| < 1,\ |b| < 1$
$\therefore \quad |ab| = |a||b| < 1$
**Now,** $a(a+b) + a^2\left(a^2+b^2\right) + a^3\left(a^3+b^3\right) + \ldots$
$= \quad \left[\left(a^2+a^4+a^6+\ldots\right)\right] + \left[\left\{ab + (ab)^2 + (ab)^3 + \ldots\right\}\right]$
$= \frac{a^2}{1-a^2} + \frac{ab}{1-ab}$

**Q.66 Answer:**

$\mathrm{f(x)}$ has local maximum at $\mathrm{x} = 1$

**Solution:**

$f(x) = (x-1)^2(x-2)^3 e^x$
$\Rightarrow \mathrm{f'(x)} = (\mathrm{x}-1)(\mathrm{x}-2)^2\left(\mathrm{x}^2+2\mathrm{x}-5\right)\mathrm{e}^{\mathrm{x}}$
$f(x)$ **has local maximum at** $x = 1$**, since** $f'(x)$ **changes from positive to negative.** $\mathrm{f(x)}$ **has a point of inflexion at** $\mathrm{x} = 2$**.**

**Q.67 Answer:**

$C + B$

**Solution:**

**Given,** $C^2 - BC + CB = 0$
$\Rightarrow C^2 - BC + CB - B^2 + B^2 = 0$
$\Rightarrow (C-B)(C+B) + B^2 = 0$
$\Rightarrow (C-B)(C+B) = I$ **(as** $B^2 = -I$ **)**
**Hence,** $(C-B)^{-1} = C + B$

**Q.68 Answer:**

$-\frac{1}{3}$

**Solution:**

**Let**
$z = \frac{x^2-6x+5}{x^2+2x+1}$
$\Rightarrow z = \frac{x^2+2x+1-8x+4}{x^2+2x+1}$
$\Rightarrow z = 1 - \frac{8x-4}{x^2+2x+1}$

$\Rightarrow z = 1 - y \ \ldots(i)$

**Now,**

$y = \frac{8x-4}{x^2+2x+1}$

$\Rightarrow \frac{dy}{dx} = \frac{8\left(x^2+2x+1\right)-(8x-4)(2x+2)}{\left(x^2+2x+1\right)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{8x^2+16x+8-16x^2-8x+8}{\left(x^2+2x+1\right)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{-8x^2+8x+16}{\left(x^2+2x+1\right)^2}$

**For maxima,** $\frac{dy}{dx} = 0$

$\Rightarrow \frac{-8x^2+8x+16}{\left(x^2+2x+1\right)^2} = 0$

$\Rightarrow x^2 - x - 2 = 0$

$\Rightarrow x = \frac{1 \pm \sqrt{1+8}}{2}$

$\Rightarrow x = 2, \ -1$

**Now,**

$y(-1) \to \infty$

$y(2) = \frac{12}{9} = \frac{4}{3}$

**Therefore, maximum value of $y$ is $\frac{4}{3}$. Therefore, minimum value of $z$ is $1 - \frac{4}{3} = -\frac{1}{3}$.**

**Q.69 Answer:**

$\mathrm{P(x)P'''(x)}$

**Solution:**

$\mathrm{y^2 = P(x) \Rightarrow 2y_1 = P'(x) \Rightarrow y_1 = \frac{P'(x)}{2y}}$

$\mathrm{\therefore y_2 = \frac{2yP''(x) - P'(x) \cdot 2y_1}{4y^2} = \frac{2yP''(x) - P'(x) \cdot \frac{P'(x)}{y}}{4y^2}}$

$\mathrm{\Rightarrow y_2 = \frac{2y^2P''(x) - \left[P'(x)\right]^2}{4y^3} = \frac{2P(x)P''(x) - \left[P'(x)\right]^2}{4y^3}}$

$\mathrm{\Rightarrow 2y^3y_2 = \frac{1}{2}\left[2P(x)P''(x) - \{P'(x)\}^2\right]}$

$\mathrm{\Rightarrow 2\frac{d}{dx}\left(y^3y_2\right) = \frac{1}{2}\left[\begin{array}{c} 2P(x)P'''(x) + 2P'(x)P''(x)] \\ -2P'(x)P''(x) \end{array}\right]}$

$\mathrm{\Rightarrow 2\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right) = P(x)P'''(x)}$

**Q.70 Answer:**

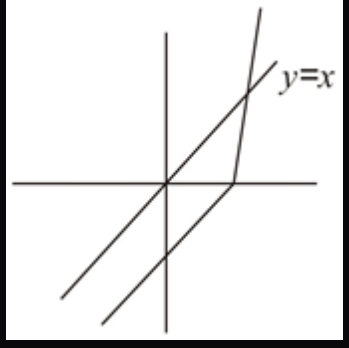
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**Solution:**

$f(x) = 2x - 1 + \left|x - \frac{1}{2}\right|$

$= \begin{cases} x - \frac{1}{2} & x < \frac{1}{2} \\ 3x - \frac{3}{2} & x \geq \frac{1}{2} \end{cases}$

**As $f(x)$ is increasing, solution of equation $f(x) = f^{-1}(x)$ is same as $f(x) = x$.**




**$\therefore$ No. of solutions of $f(x) = f^{-1}(x)$ is one.**

**Q.71 Answer:**

—

**Solution:**

**We know that,** $|x| = \begin{cases} +x, & \text{when } x \geq 0 \\ -x, & \text{when } x < 0 \end{cases}$

**For,** $x \geq 0$,

$f(x) = (x - x)^2(1 - x + x)^2 = 0$

**And, for** $x < 0$,

$f(x) = (x - (-x))^2(1 - x + (-x))^2$

$= (x + x)^2(1 - 2x)^2 = 4x^2\left(1 + 4x^2 - 4x\right)$

$=16x^4 - 16x^3 + 4x^2$

$\Rightarrow f\left(x\right) = 16x^4 - 16x^3 + 4x^2$

**We have** $f(x) = \begin{cases} 16x^4 - 16x^3 + 4x^2, & x < 0 \\ 0, & x \geq 0 \end{cases}$

**Now, Left hand Limit,** LHL **at** $x = 0$

$= \lim\limits_{x \to 0^-} f(x) = \lim\limits_{h \to 0} f(0 - h) = \lim\limits_{h \to 0}\left(16h^4 - 16h^3 + 4h^2\right) = 0$

**And Right hand limit,** RHL **at** $x = 0$

$= \lim\limits_{x \to 0^+} f(x) = \lim\limits_{h \to 0} f(0 + h) = \lim\limits_{h \to 0} 0 = 0$

**And,** $f(0) = 0$

**Therefore,** LHL **at** $(x = 0)$ = RHL **at** $(x = 0) = f(0)$

**Hence, the function** $f(x)$ **is continuous at** $x = 0$

**For,** $x > 0, f(x) = 0$, **So,** $f(x)$ **is continuous for all** $x > 0$.

**And for** $x < 0, f(x)$ **is a polynomial function of degree** $4$ **which is always continuous for all Real value of** $x$.

**Therefore** $f(x)$ **is continuous for all** $x < 0$.

**Hence** $f(x)$ **is continuous for all** $x \in R$.

$f'(x) = \begin{cases} 64x^3 - 48x^2 + 8x, x < 0 \\ 0, x \geq 0 \end{cases}$

$\lim\limits_{x \to 0^-} f'(x) = \lim\limits_{h \to 0} f'(0 - h) = \lim\limits_{h \to 0}\left(64h^3 - 48h^2 + 8h\right) = 0$

**And,** $\lim\limits_{x \to 0^+} f'(x) = \lim\limits_{h \to 0} f'(0 + h) = \lim\limits_{h \to 0} 0 = 0$

**Therefore, Left hand derivative at** $(x = 0)$ =**Right hand derivative at** $(x = 0) = 0$

**Therefore,** $f(x)$ **is differentiable at** $x = 0$.

**For,** $x > 0, f'(x) = 0$ **for all values of** $x$. **So,** $f(x)$ **is differentiable for all** $x > 0$.

**For,** $x < 0, f'(x)$ **is a polynomial function of degree** $4$ **which is differentiable for all values** $x$..

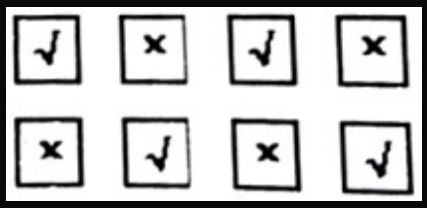
Hence, $f(x)$ is continuous and differentiable for all $x \in R$. So, number of points in the interval $(-3, 4)$ where $f(x)$ is not differentiable is Zero.

**Q.72 Answer:**

—

**Solution:**

Clearly guests will stay either in '✓' or in '×'



Therefore, number of required ways $= 2 \times 4! = 48$ .

**Q.73 Answer:**

—

**Solution:**

Given that

$x\ dy = y\ dx + y^2 dy$

$\int \frac{ydx - xdy}{y^2} = \int -dy$

We know that $d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$

$\frac{x}{y} = -y + c$

As, $y(1) = 1$

$1 = -1 + c$

$\Rightarrow c = 2$

$\Rightarrow \frac{x}{y} = -y + 2$

If $x = -3$

$-3 = -y^2 + 2y$

$\Rightarrow y^2 - 2y - 3 = 0$

$\Rightarrow y = 3$

**Q.74 Answer:**

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**Solution:**

We have,

$A = \{1, 2, 3, 4\}$

Now, $B \subseteq A,\ C \subseteq A$ and $B \cap C = \phi$.

Number of such ordered pairs are $3^{n(A)} = 3^4$

**Q.75 Answer:**

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**Solution:**

**It is given that,**

$\mathrm{A} = \begin{bmatrix} \frac{-1+i\sqrt{3}}{2i} & \frac{-1-i\sqrt{3}}{2i} \\ \frac{1+i\sqrt{3}}{2i} & \frac{1-i\sqrt{3}}{2i} \end{bmatrix},\ i = \sqrt{-i}$

**and** $f(x) = x^2 + 2$

**since,** $\omega^3 = 1$

$\omega = \frac{-1+i\sqrt{3}}{2}$ **and** $\omega^2 = \frac{-1-i\sqrt{3}}{2}$

**and** $\omega^2 + \omega = -1$

**Then,** $\mathrm{A} = \begin{bmatrix} \frac{\omega}{i} & \frac{\omega^2}{i} \\ \frac{-\omega^2}{i} & -\frac{\omega}{i} \end{bmatrix}$

$= \frac{\omega}{i}\begin{bmatrix} 1 & \omega \\ -\omega & -1 \end{bmatrix}$

$\mathrm{A}^2 = \frac{\omega^2}{i^2}\begin{bmatrix} 1 & \omega \\ -\omega & -1 \end{bmatrix}\begin{bmatrix} 1 & \omega \\ -\omega & -1 \end{bmatrix}$

$= -\omega^2\begin{bmatrix} 1-\omega^2 & 0 \\ 0 & 1-\omega^2 \end{bmatrix}$

$= \begin{bmatrix} -\omega^2+\omega^4 & 0 \\ 0 & -\omega^2+\omega^4 \end{bmatrix}$

$= \begin{bmatrix} -\omega^2+\omega & 0 \\ 0 & -\omega^2+\omega \end{bmatrix}$

$\therefore\ f(x) = x^2 + 2$

$\Rightarrow f(\mathrm{A}) = \mathrm{A}^2 + 2$

$= \begin{bmatrix} -\omega^2+\omega & 0 \\ 0 & -\omega^2+\omega \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} -\omega^2+\omega+2 & 0 \\ 0 & -\omega^2+\omega+2 \end{bmatrix}$

$= (-\omega^2+\omega+2)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \left(3+2\omega\right)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\{\because\ 1+\omega+\omega^2 = 0\}$

$= \left[3+2\left(\frac{-1+i\sqrt{3}}{2}\right)\right]\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \left(2+i\sqrt{3}\right)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\therefore\ \left|\frac{f(\mathrm{A})}{2+i\sqrt{3}}\right| = 1$